



MAGNETICUM



Numerical Simulations of physical processes driving galaxy evolution

Lecture 3: Simulation Types

Rhea-Silvia Remus

Canary Islands Winter School, 26.11.2021



Disclaimer

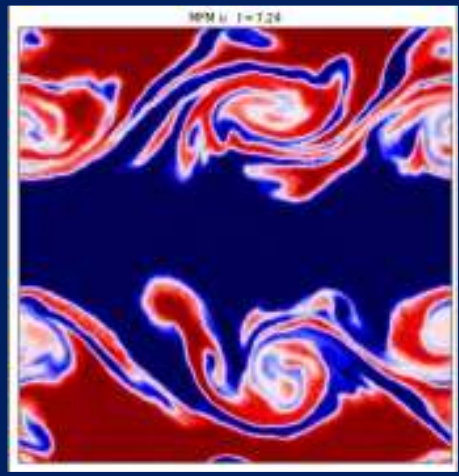
This lecture was designed for presentation with movie media. If no specific URL is provided, movies can be found at

www.usm.uni-muenchen.de/~rhea/teaching/movies

Movies are marked by a *

Summary: Computational Methods

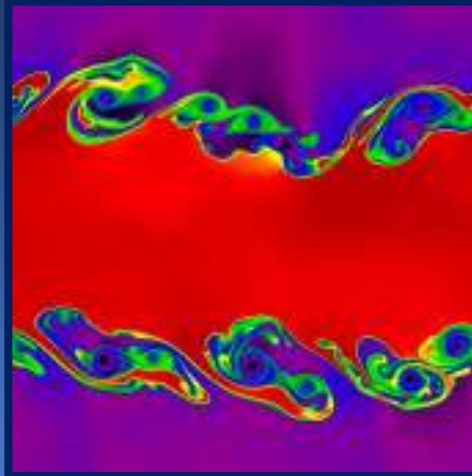
Smooth Particle Hydrodyn.



Courtesy
M. Niemeyer, K. Dolag

- ✓ Very good conservation properties (mass, momentum, total energy, angular momentum, entropy)
- ✓ shape invariant
- Instabilities do not grow sufficiently
- Mixing behind shocks not sufficient
- Shocks captured by artificial viscosity

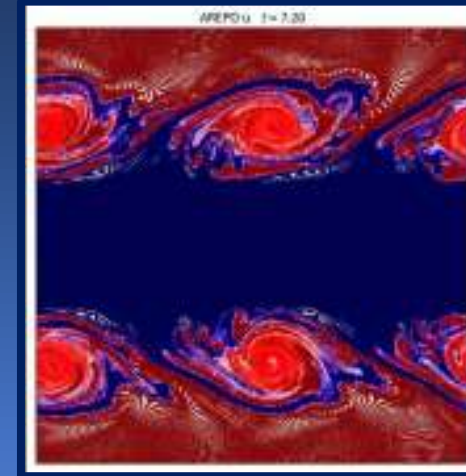
Adaptive Mesh Refinement



<https://www.astro.princeton.edu/~jstone/Athena/tests/kh/kh.html>

- ✓ Instabilities nicely grow
- ✓ Mixing between phases works well
- Energy conservation issues (especially for fast moving elements)
- Flow over cell boundaries becomes an issue for adaptive meshes
- Not shape invariant

Moving Mesh

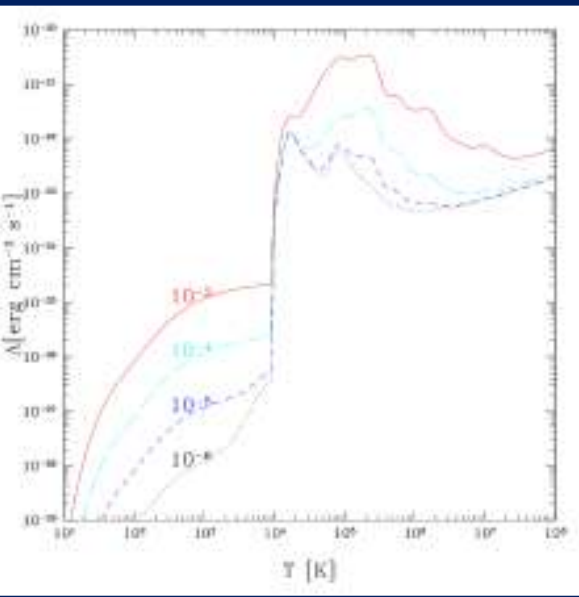


Courtesy
M. Niemeyer, K. Dolag

- ✓ All good things from the other two
- Flow over cell boundaries (only pseudo-Lagrangian)

Summary: Including Physics

Cooling



Maio et al.,
2007

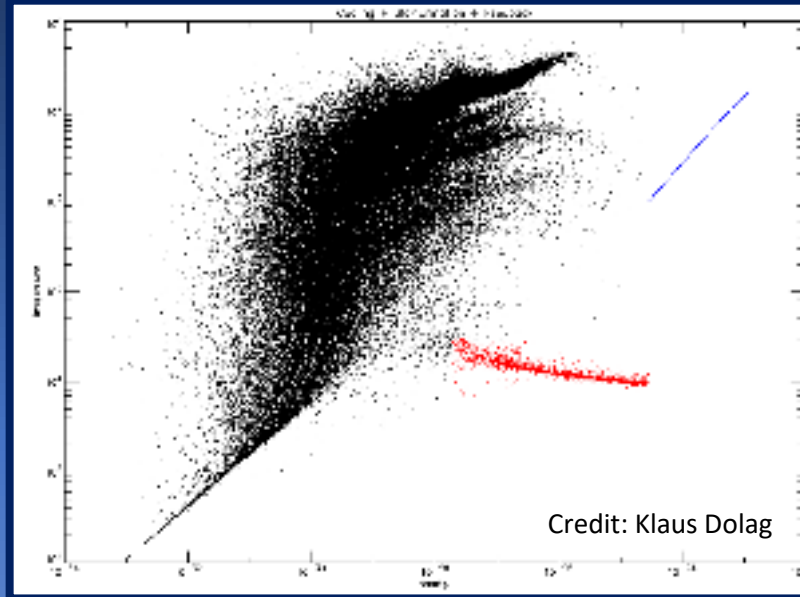
Basic Assumption:

- Optically thin
- Ionization equilibrium (H , H^+ , He , He^+ , He^{++} , e^-)
- 2-body processes ($\sim n^2$)

$$\Lambda(T)/n^2$$

BUT: Cooling Catastrophe

Star Formation



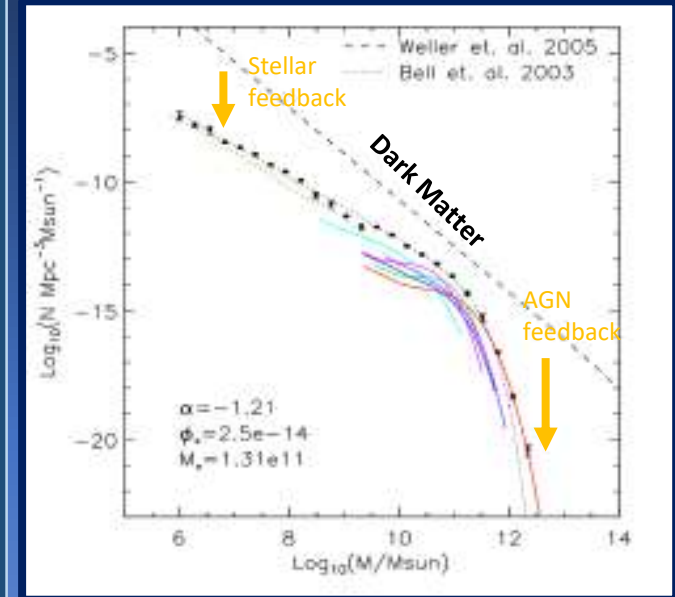
Credit: Klaus Dolag

Star formation subgrid model:

- Self-regulated star formation
- Set of differential equations needs to be solved.
- Produces reasonable galaxies at low z

BUT: star formation rates at high z not captured

Feedback



Read &
Trentham
2005

Feedback comes from two different sources:

- Massive Stars and Supernovae
- Supermassive Black Holes (AGN)

Stops the Overcooling Catastrophe

BUT: Burns holes into disks

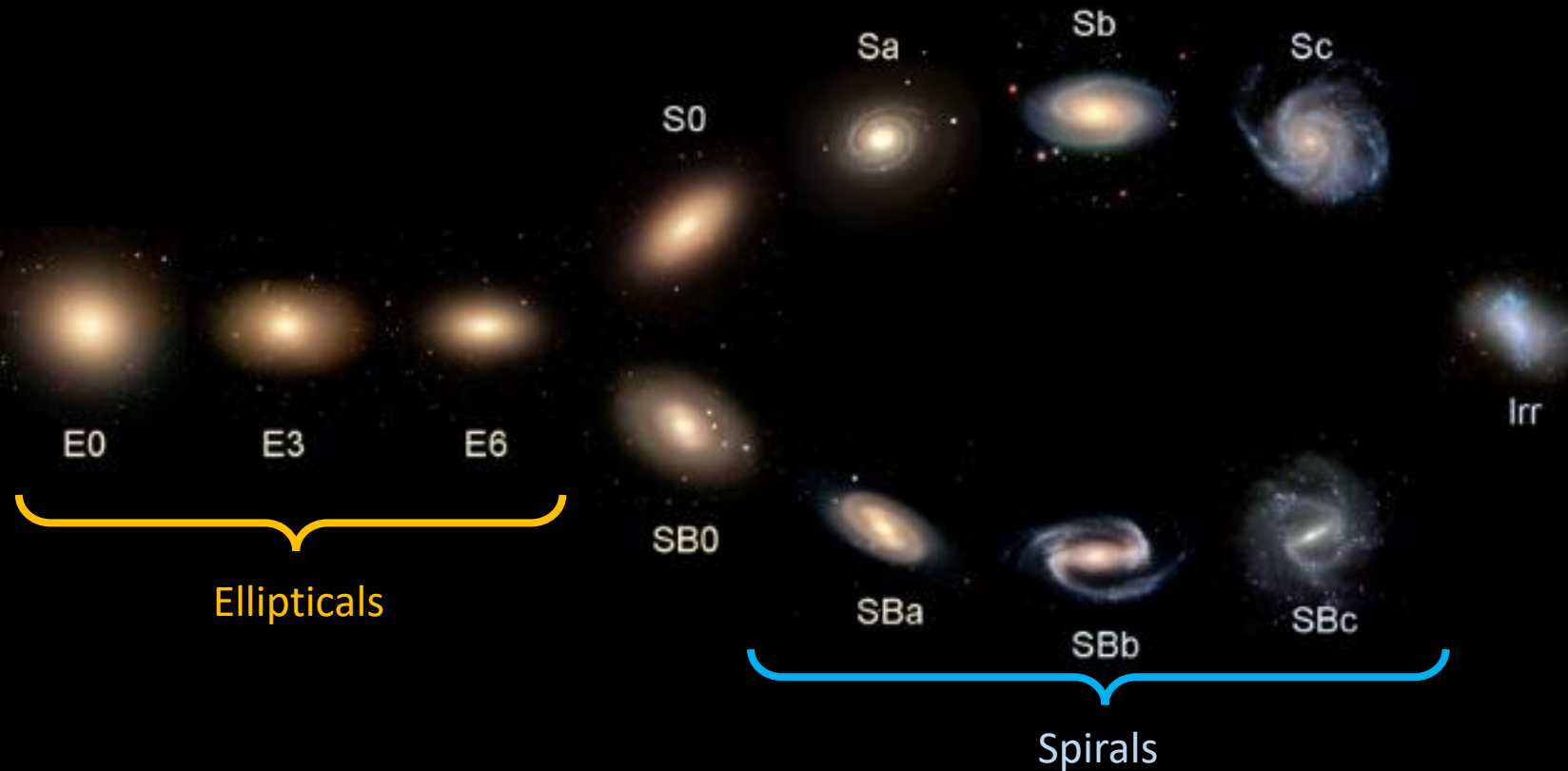


Making Galaxies (and (Proto)-Clusters)



Introduction: why do we care about galaxies?

Galaxies come in many different flavours, not just the well known regular shapes but a multitude of distorted features that need to be explained



NASA, ESA & The Hubble Heritage Team (STScI/AURA)

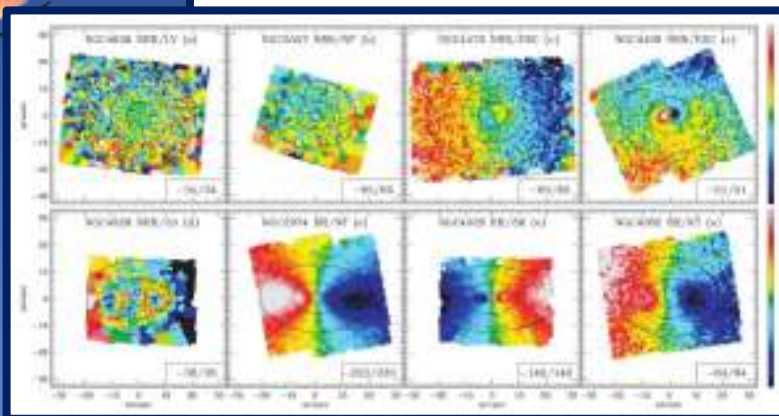
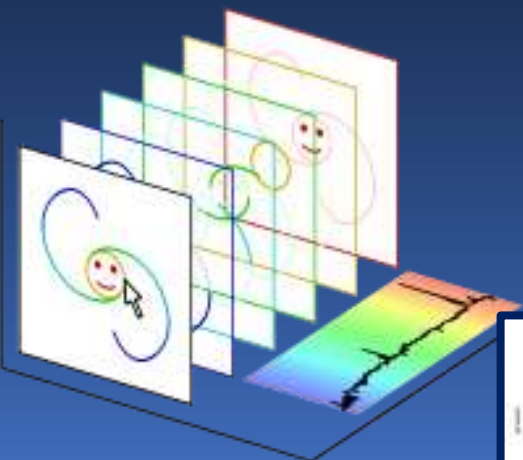


ESO/VMC Survey

P.-A. Duc (CEA, CFHT) & Atlas3D

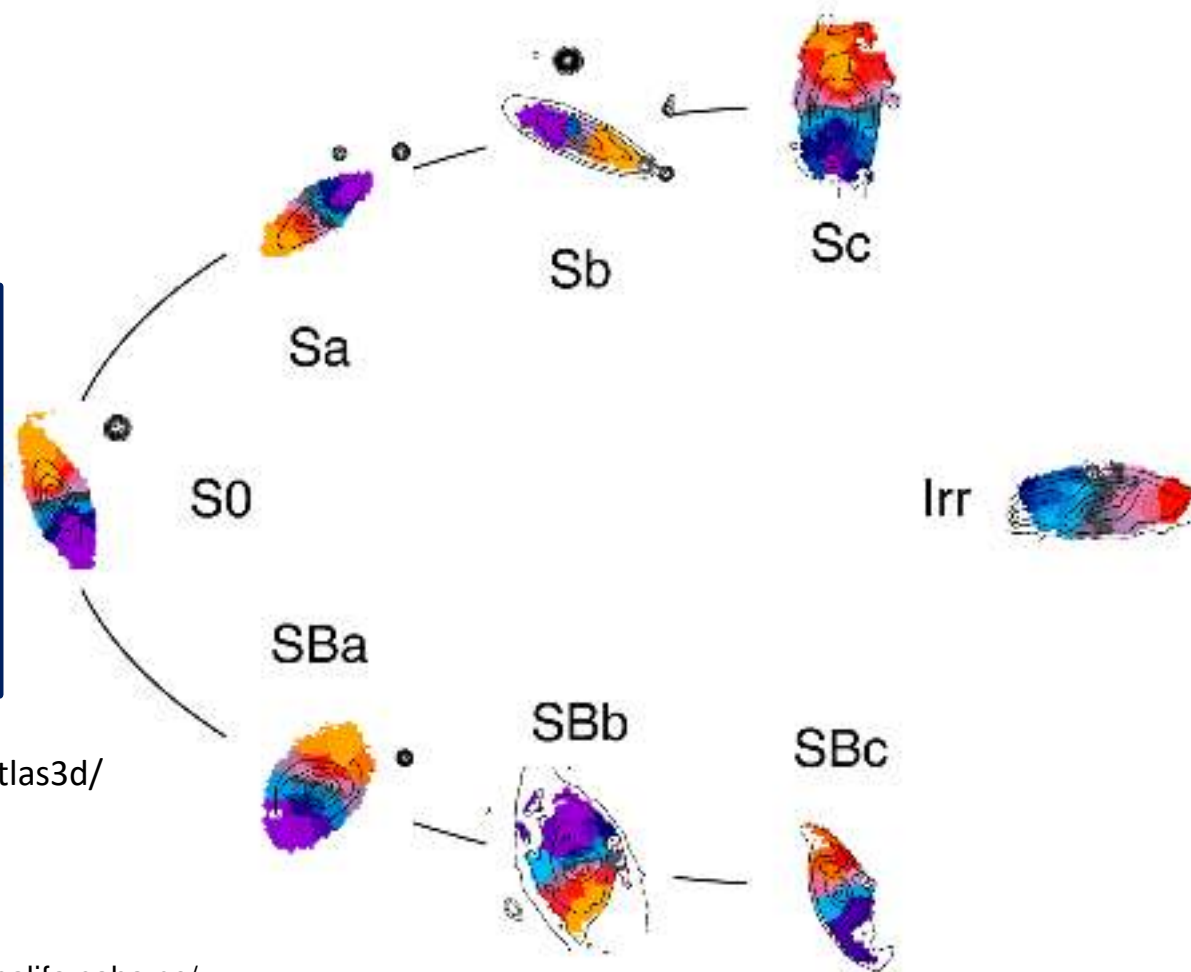
NASA, ESA & The Hubble Heritage Team

Introduction: why do we care about galaxies?



Atlas^{3D} Survey:
<https://www-astro.physics.ox.ac.uk/atlas3d/>

Pictures from the CALIFA survey: <http://califa.caha.es/>



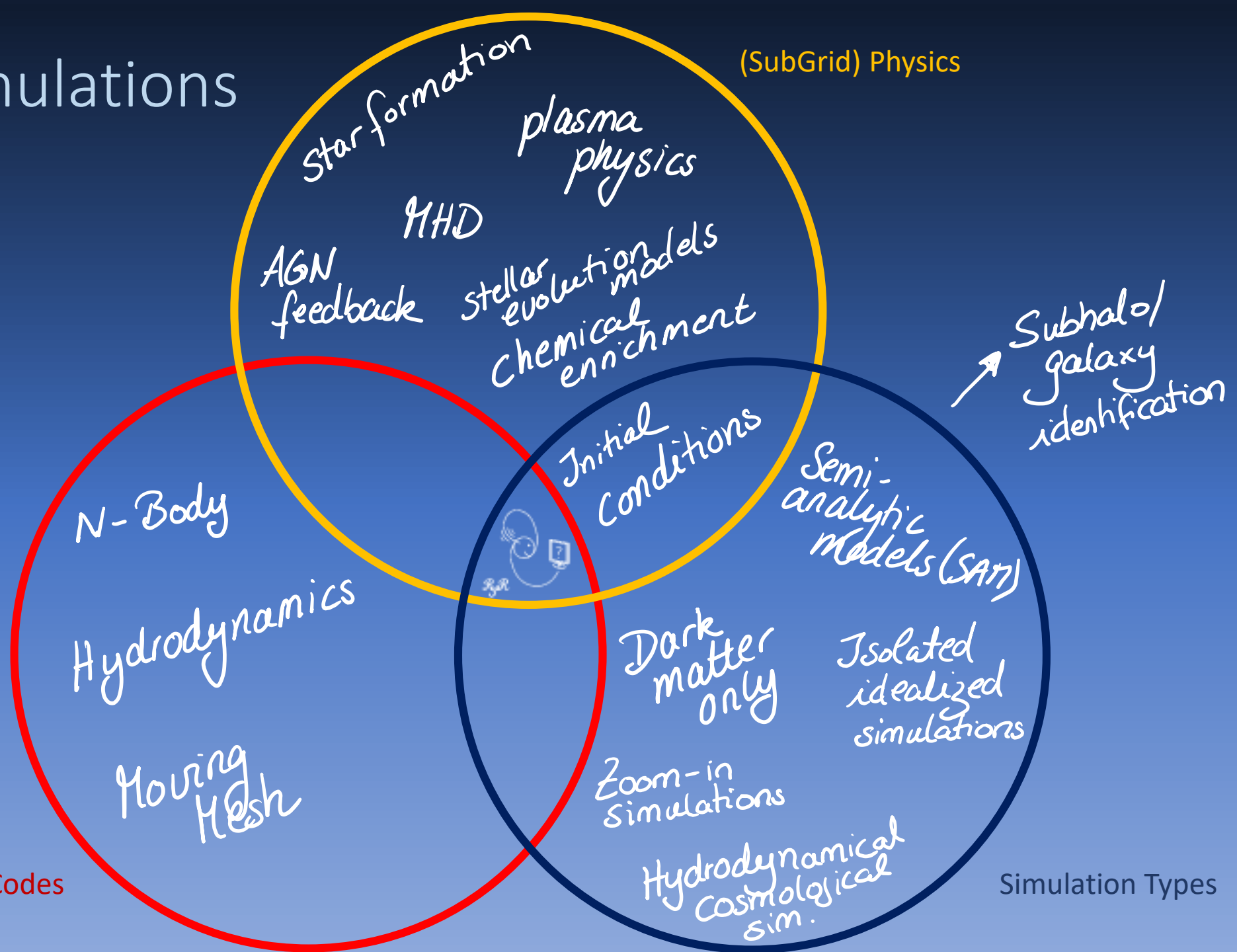
With the advent of Integral Field Spectroscopy, the features known to exist in galaxies were multiplied especially in the realm of quiescent galaxies

Numerical Simulations

Backbone Codes

(SubGrid) Physics

Simulation Types





Quiz

When was the first simulation of two merging galaxies performed?

1. 1941
2. 1972
3. 1985
4. 2001



Quiz

When was the first simulation of two merging galaxies performed?

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2. 1972
3. 1985
4. 2001

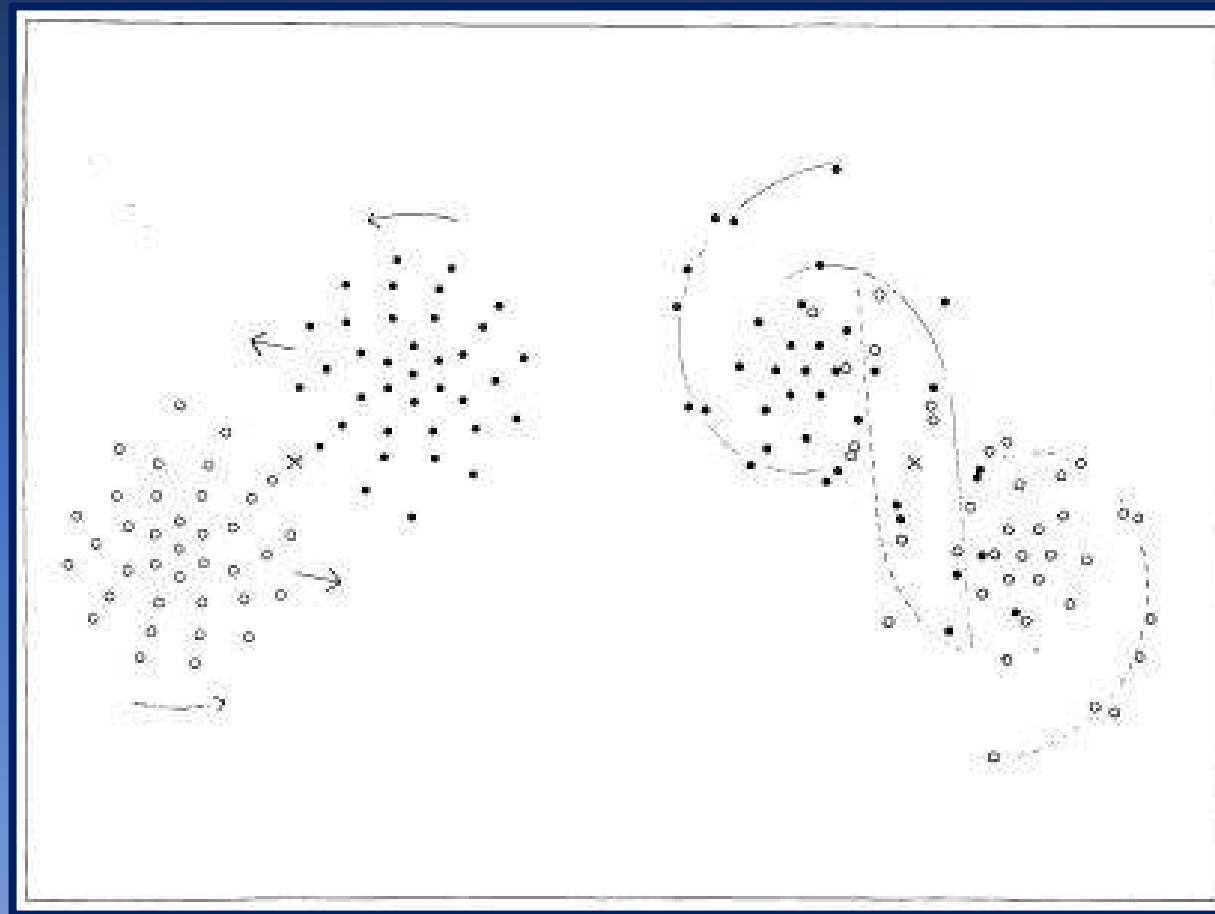


First Simulation

1941: Erik Holmberg performed a merger of two galaxies using Light Bulbs. He calculated the acceleration by integrating the light at a given position with a photocell.

Gravity: Acceleration $a = \frac{G M}{r^2} \propto \frac{1}{r^2}$

Light: Flux $F = \frac{L}{4\pi r^2} \propto \frac{1}{r^2}$

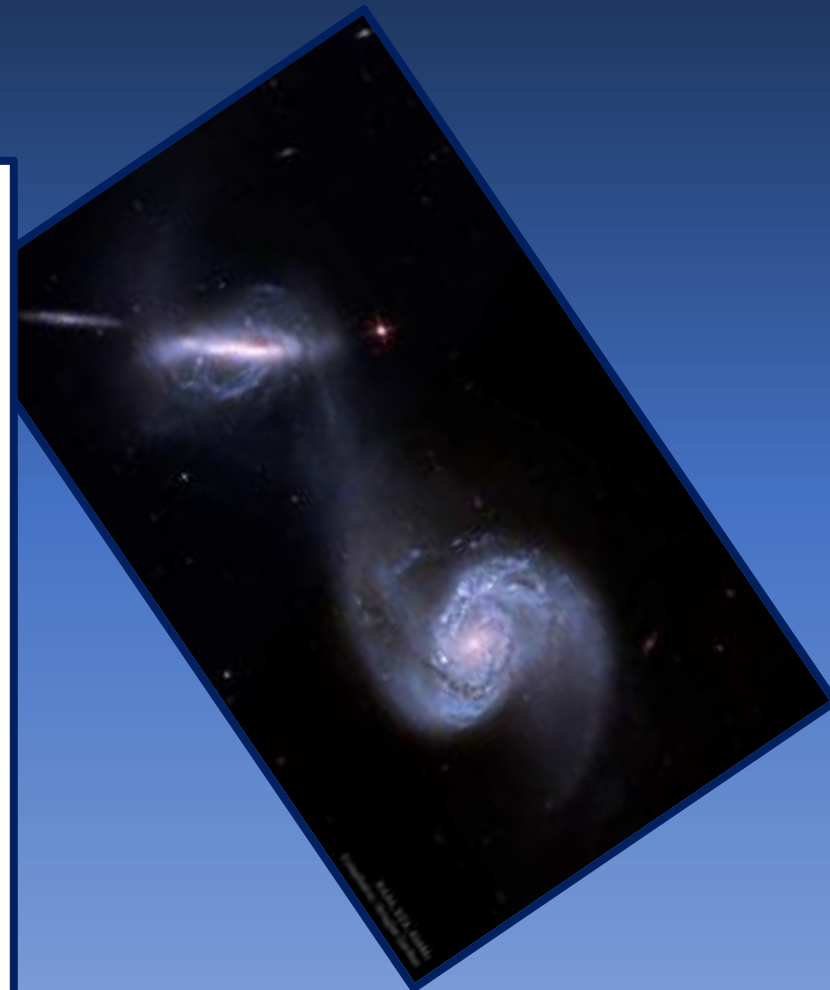
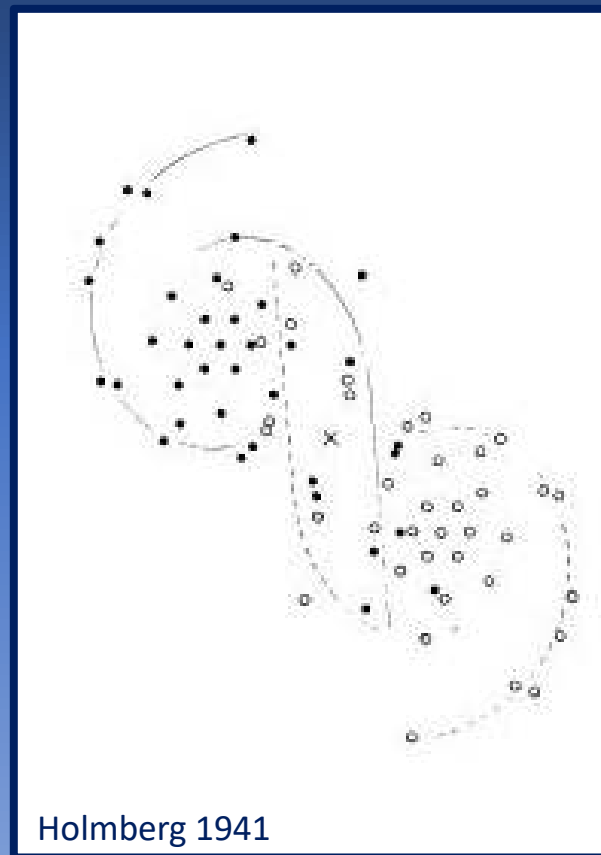
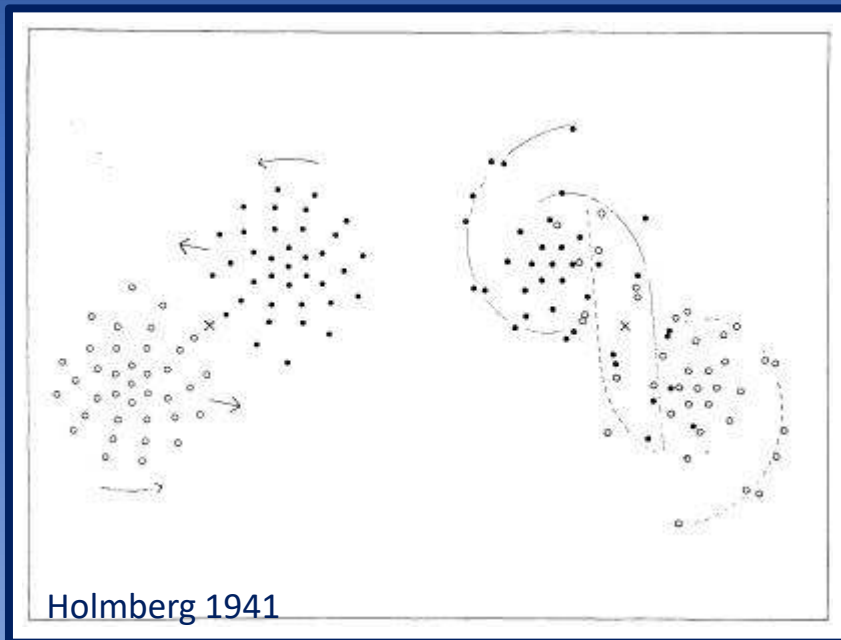


Holmberg 1941



First Simulation

1941: Erik Holmberg performed a merger of two galaxies using Light Bulbs. He calculated the acceleration by integrating the light at a given position with a photocell





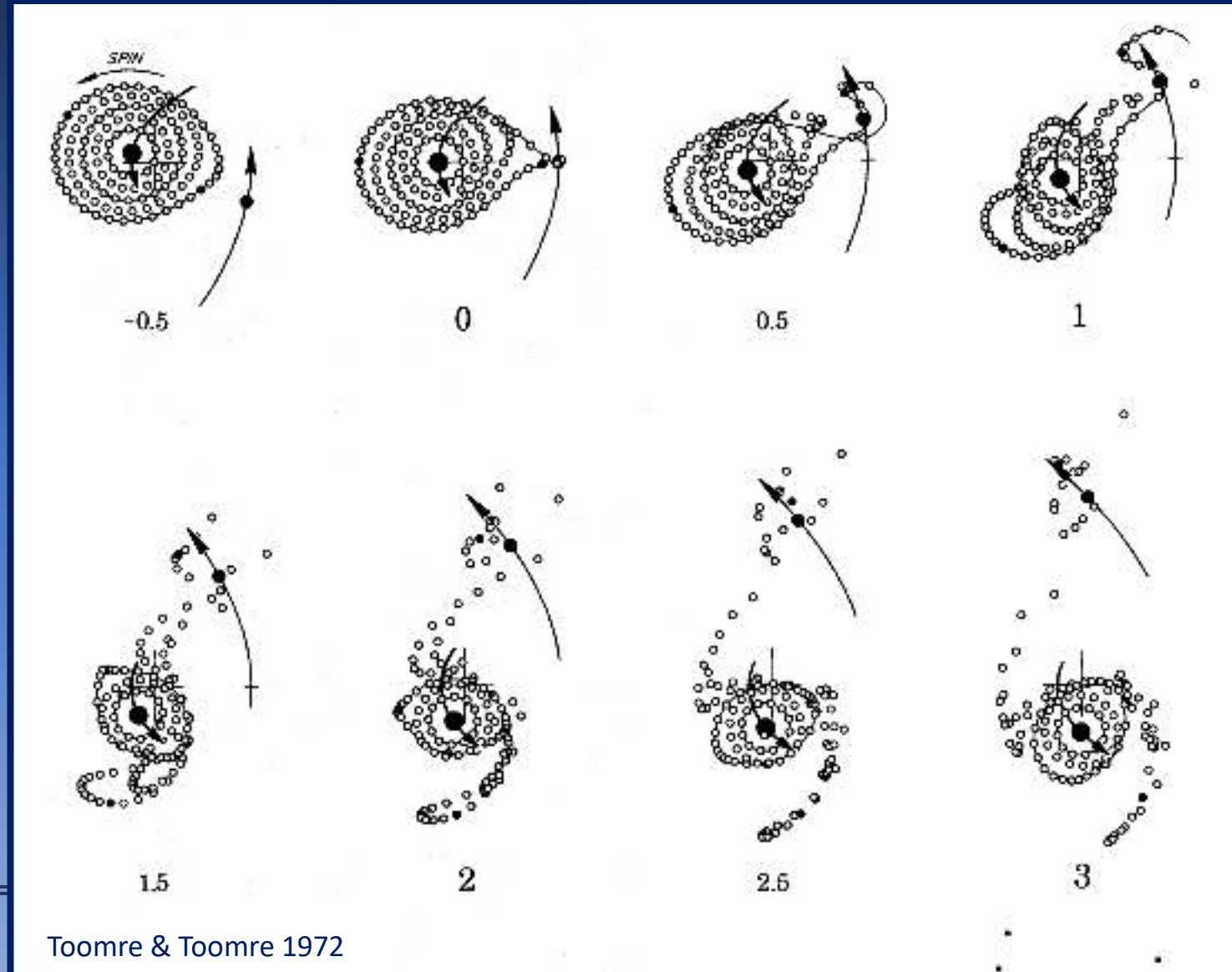
Next one: Toomre and Toomre

30 years later....

Galactic Bridges and Tails

by Toomre & Toomre, 1972

A simple fly-by already can make
arms and tidal tails



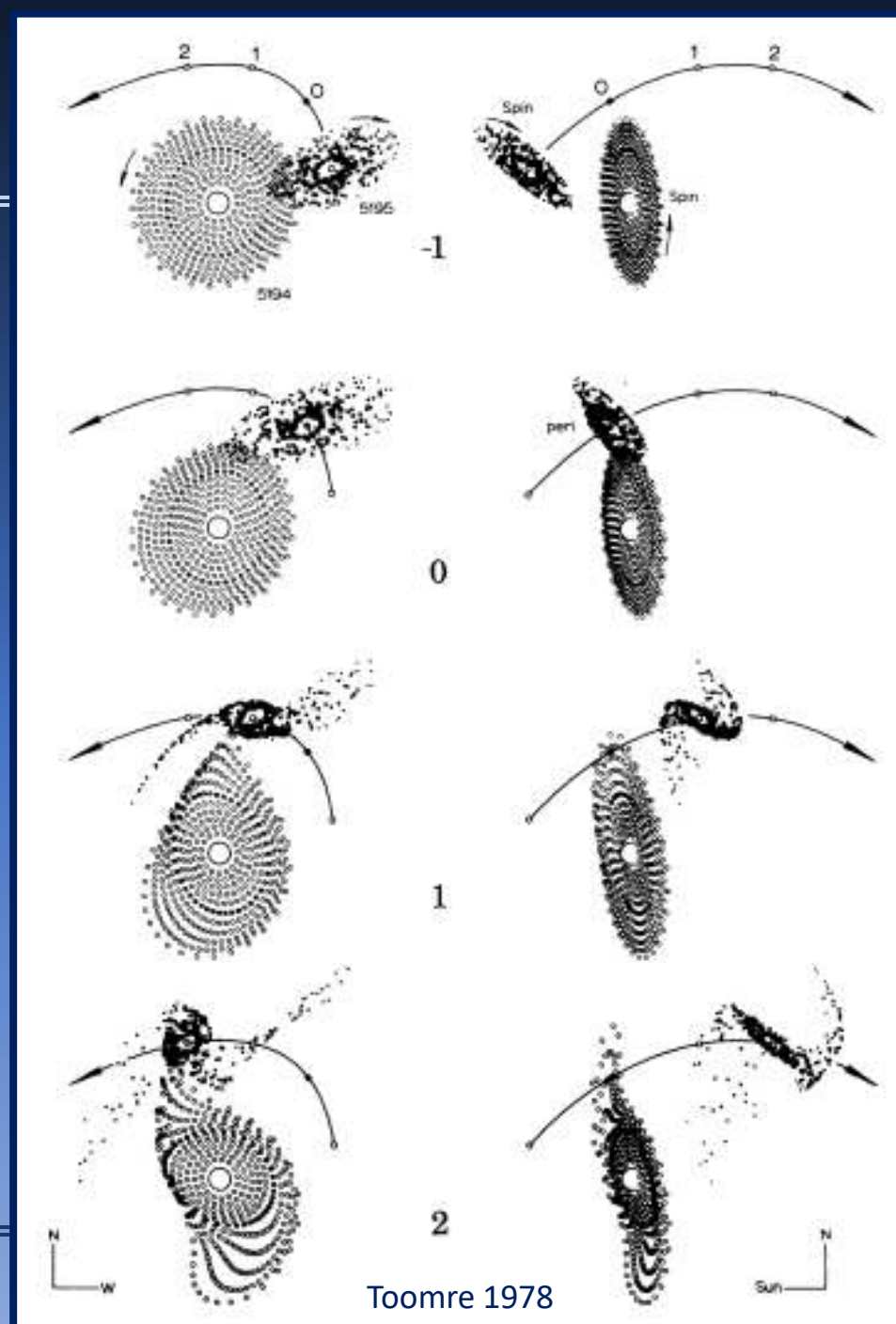


N-Body Simulations

30 years later....

Galactic Bridges and Tails

by Toomre & Toomre, 1972





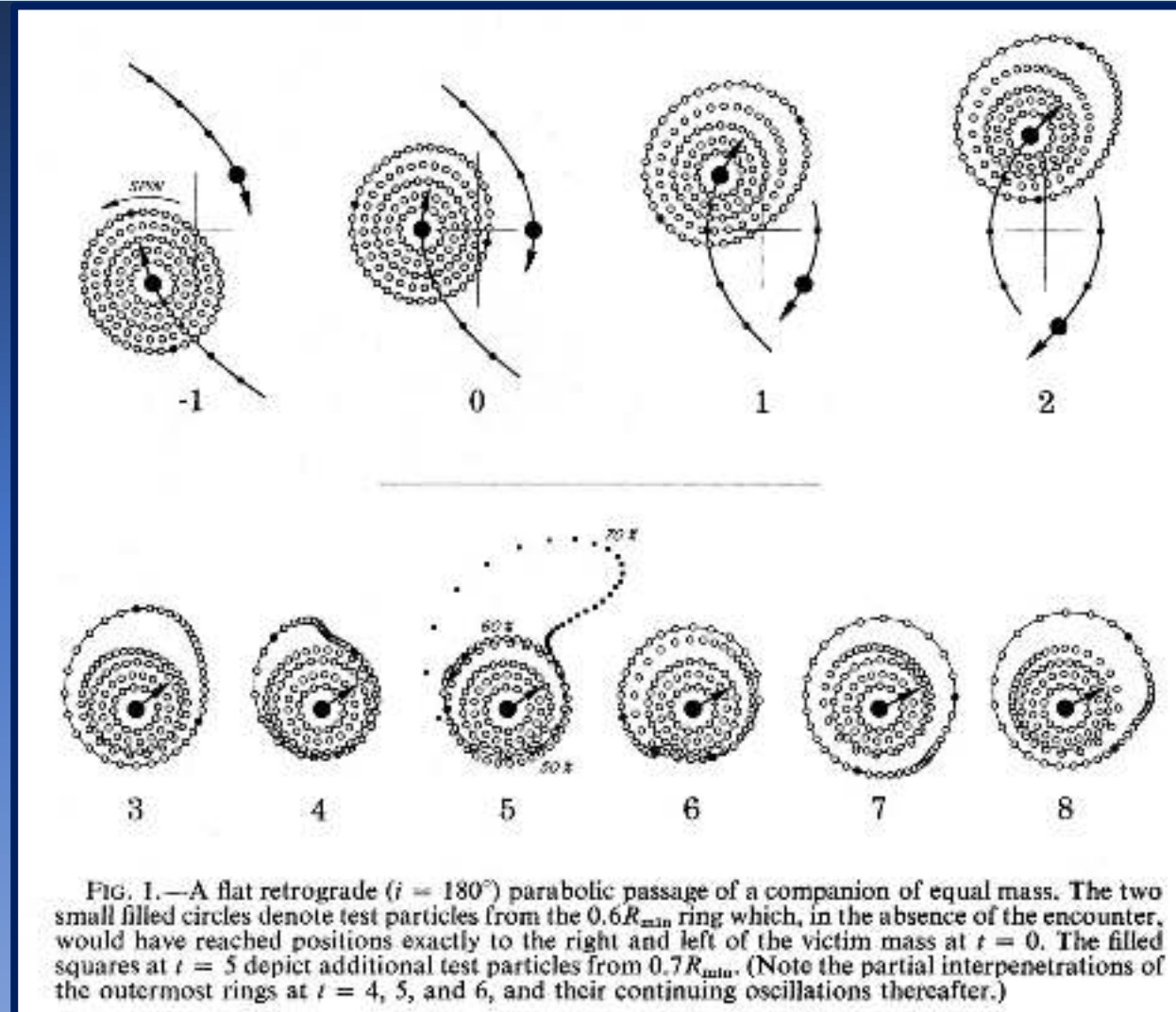
N-Body Simulations

30 years later....

Galactic Bridges and Tails

by Toomre & Toomre, 1972

N-Body Simulations





Overview: Galaxy Formation Simulations

Isolated Merger Simulations



Cosmological Zoom-Simulations



Cosmological Box Simulations



Isolated (Binary Merger) Simulations

Artificially set up galaxies with defined mass distributions.

Set two (or more) on a collision orbit with clearly defined orbital parameters (controlled initial conditions).



Isolated (Binary Merger) Simulations

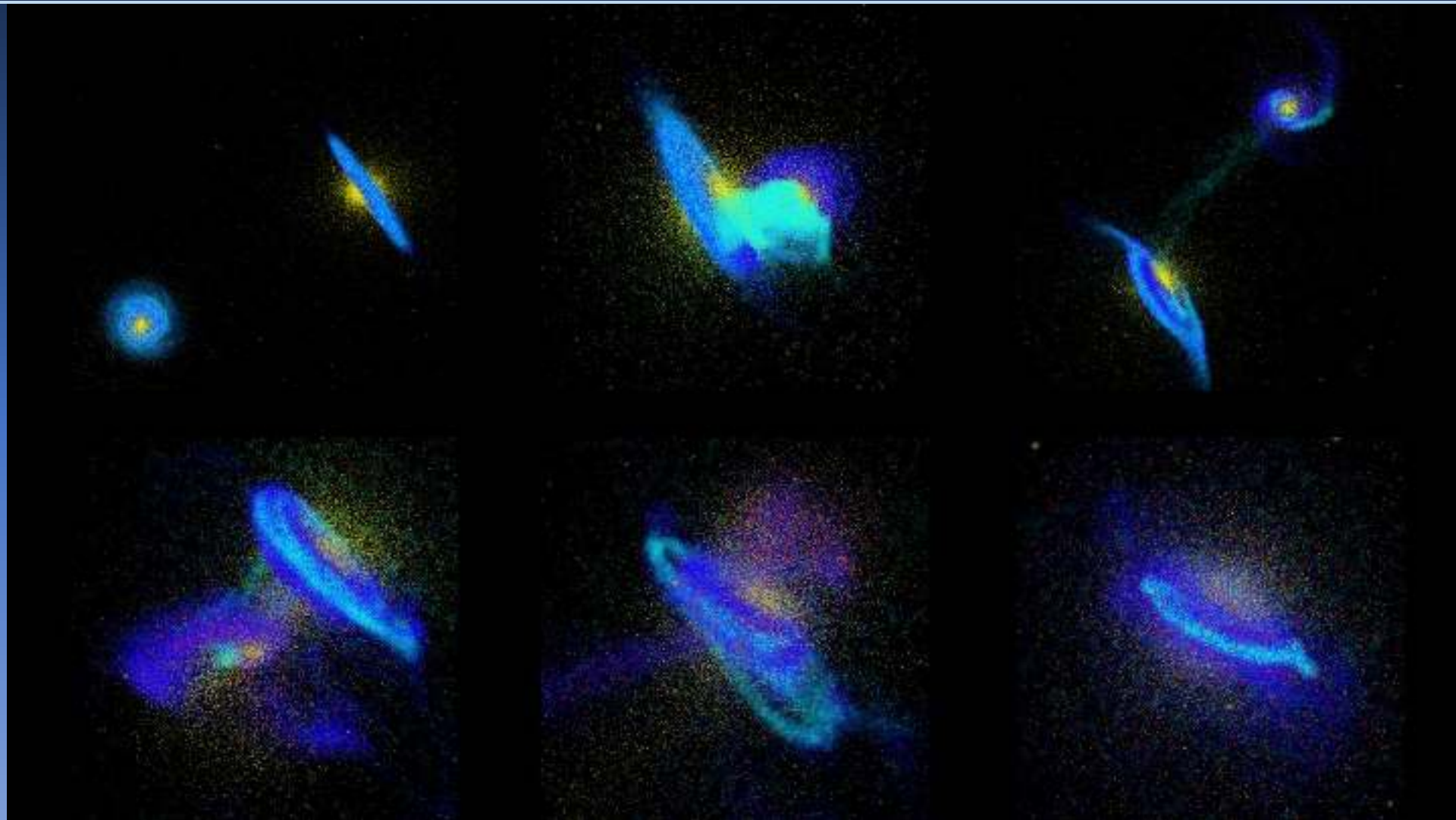


Image: Rhea-Silvia Remus

Isolated (Binary Merger) Simulations

Artificially set up galaxies
with defined mass
distributions.

Set two (or more) on a
collision orbit with clearly
defined orbital parameters



Credit: Credit: Frank Summers

<http://www.tapir.caltech.edu/~phopkins/Site/animations/Mergers/sims-versus-observations.html>

Evolution of simulating the Antennae Galaxies



Image Data: Subaru, NAOJ, NASA/ESA/Hubble, R.W. Olsen - Processing: Federico Pelliccia and Rolf Wahl Olsen (APOD)



Toomre & Toomes 1972



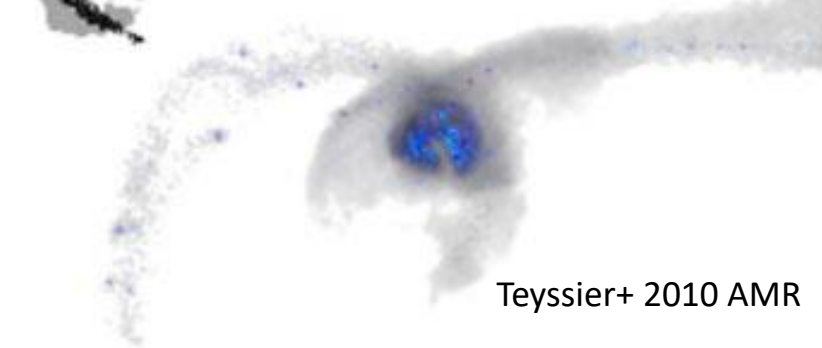
Barnes 1988



Mihos+ 1993



Karl+ 2010

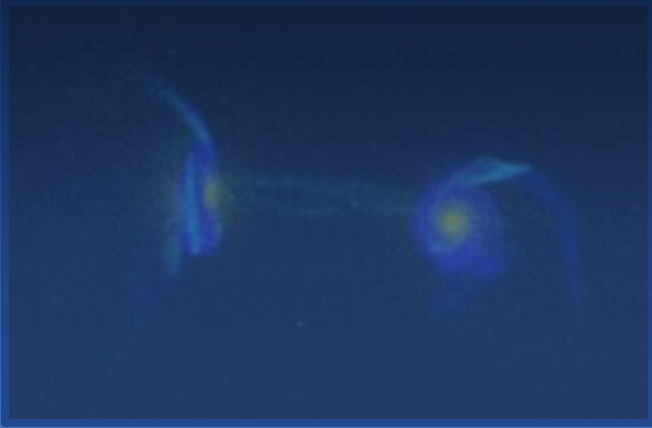


Teyssier+ 2010 AMR



Overview: Galaxy Formation Simulations

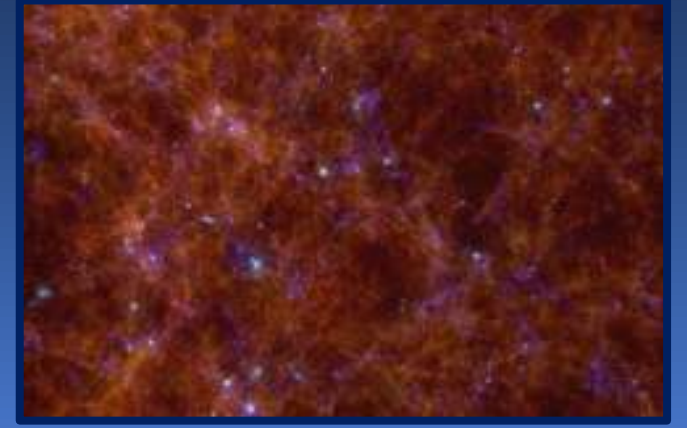
Isolated Merger Simulations



Cosmological Zoom-Simulations



Cosmological Box Simulations

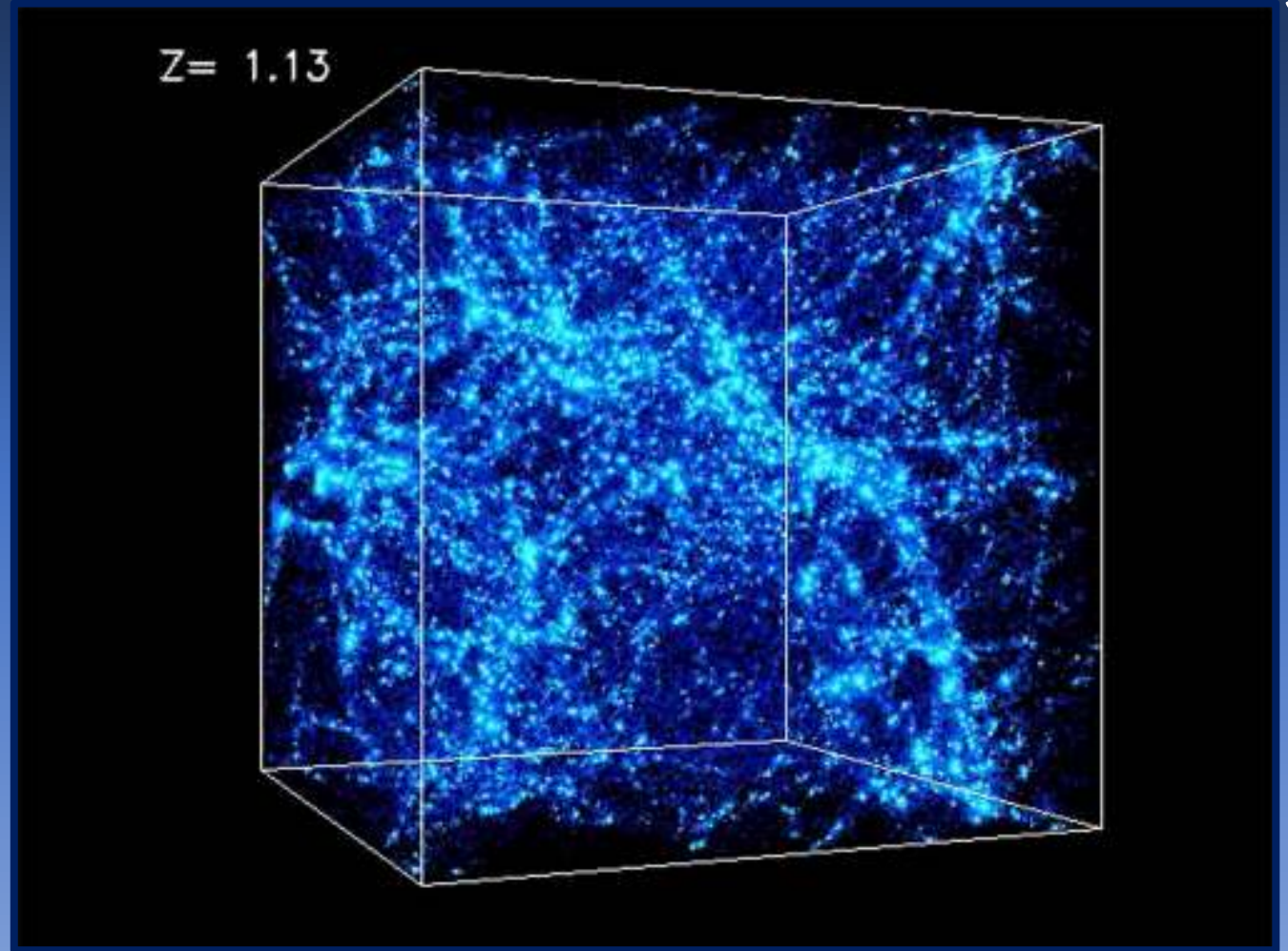


Cosmological Simulations

Simulate a full universe starting with dark matter and gas in a cubic volume.

Galaxies form due to initial perturbations – you have to take what you get.

Simulations were performed at the National Centre for Supercomputer Applications by Andrey Kravtsov (The University of Chicago) and Anatoly Klypin (New Mexico State University). Visualizations by Andrey Kravtsov.

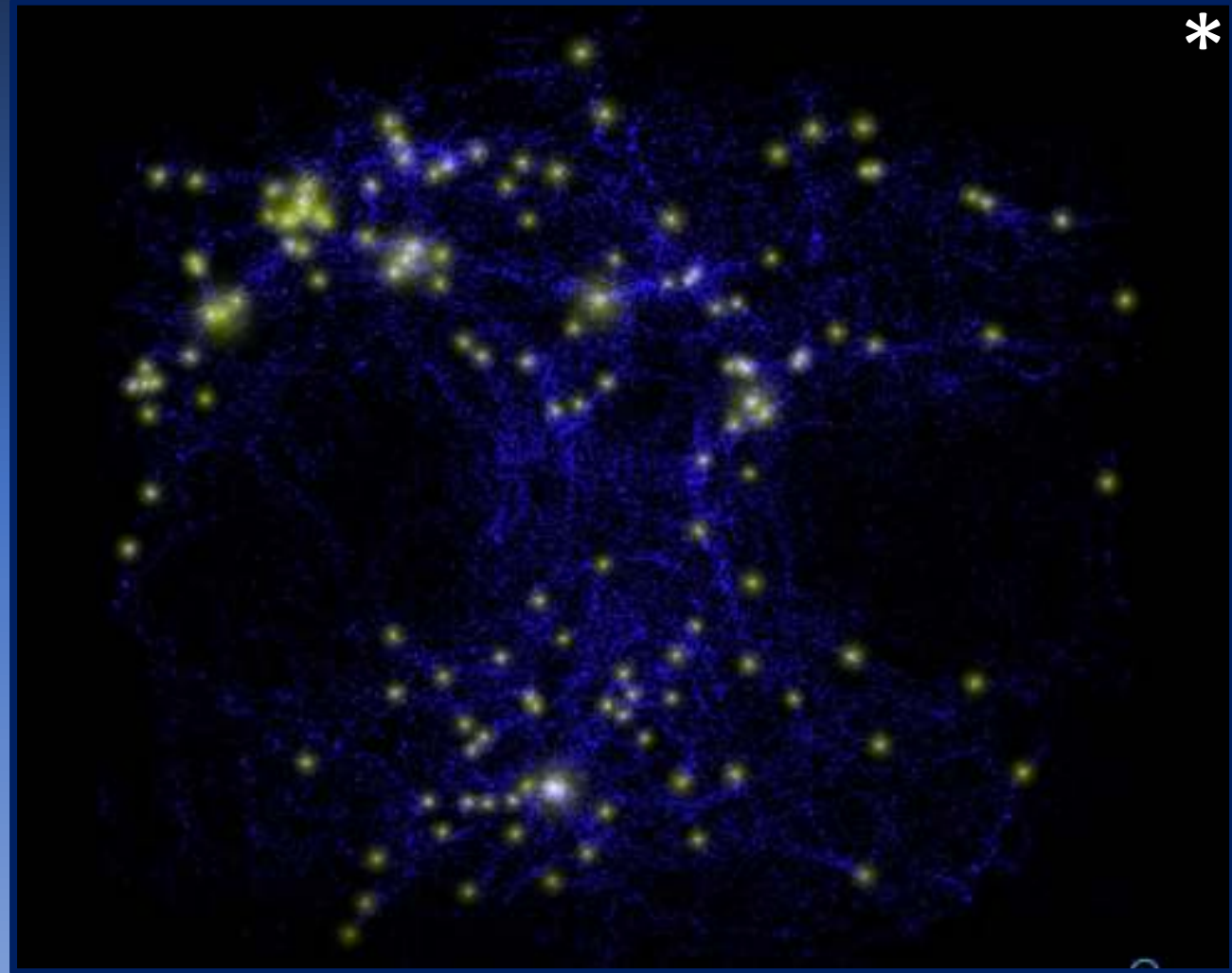


Cosmological Simulations

The most massive structures are formed in the crossings of filaments

Note: small boxes always show a cross-like structure in the cosmic web due to the largest mode in growth

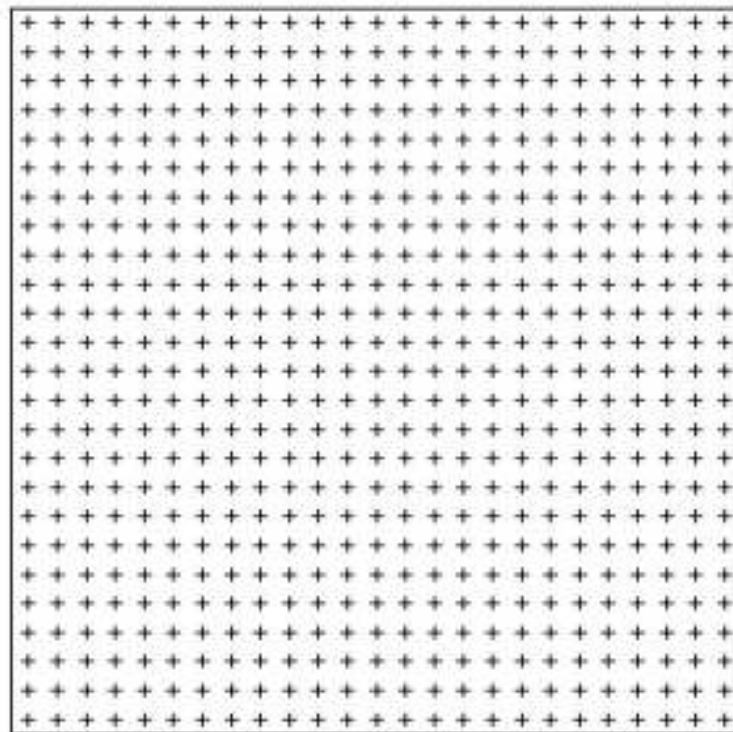
Movie credit: Rhea-Silvia Remus



Cosmological Simulations

Initial Conditions:

Box volume with periodic boundary conditions, particles set up homogeneous and isotropic.



homogeneous
&
isotropic

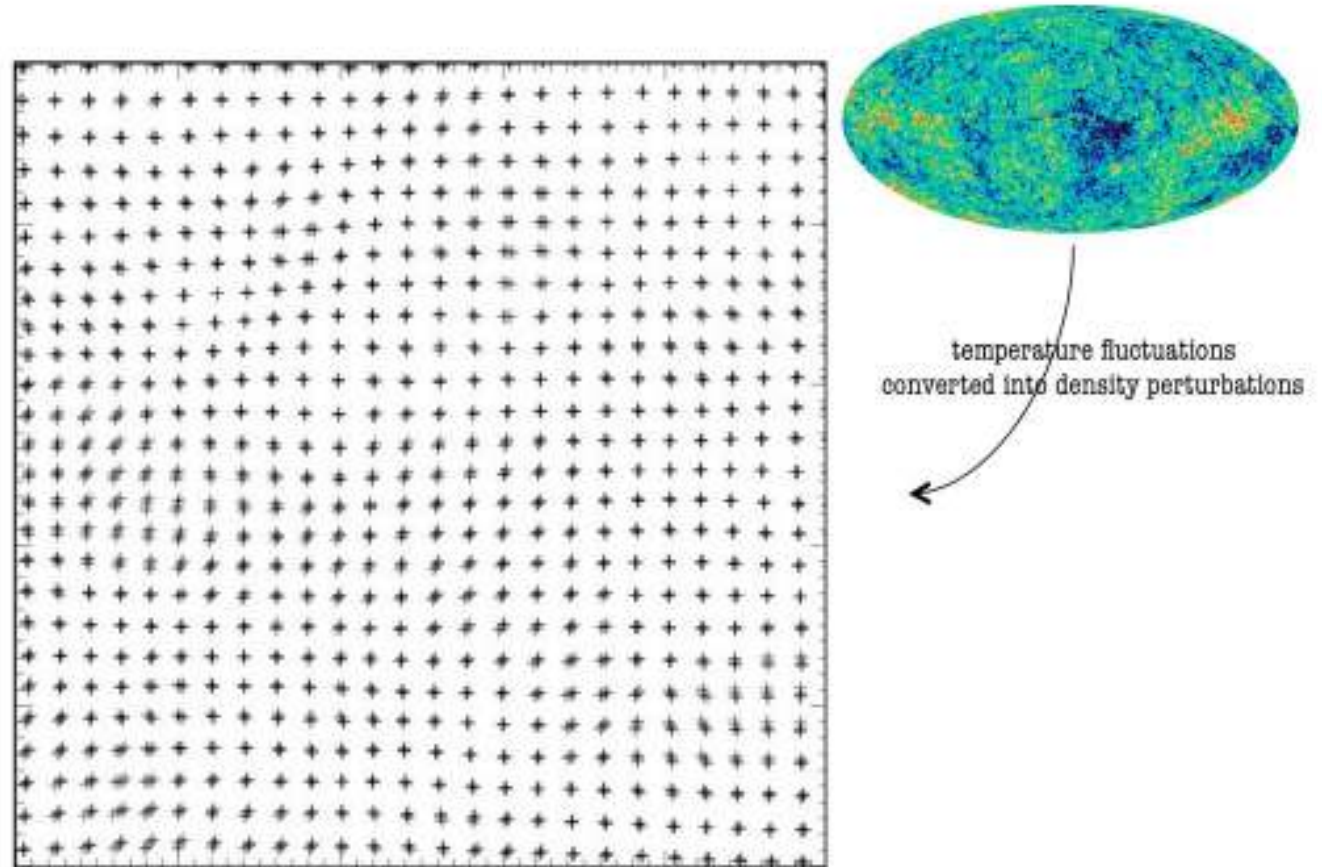
From A. Knebe: <http://popia.ft.uam.es/aknebe/page3/files/ComputationalCosmology/05ICs.pdf>

Cosmological Simulations

Initial Conditions:

Box volume with periodic boundary conditions, particles set up homogeneous and isotropic.

Then perturb the initial distribution due to chosen cosmology.



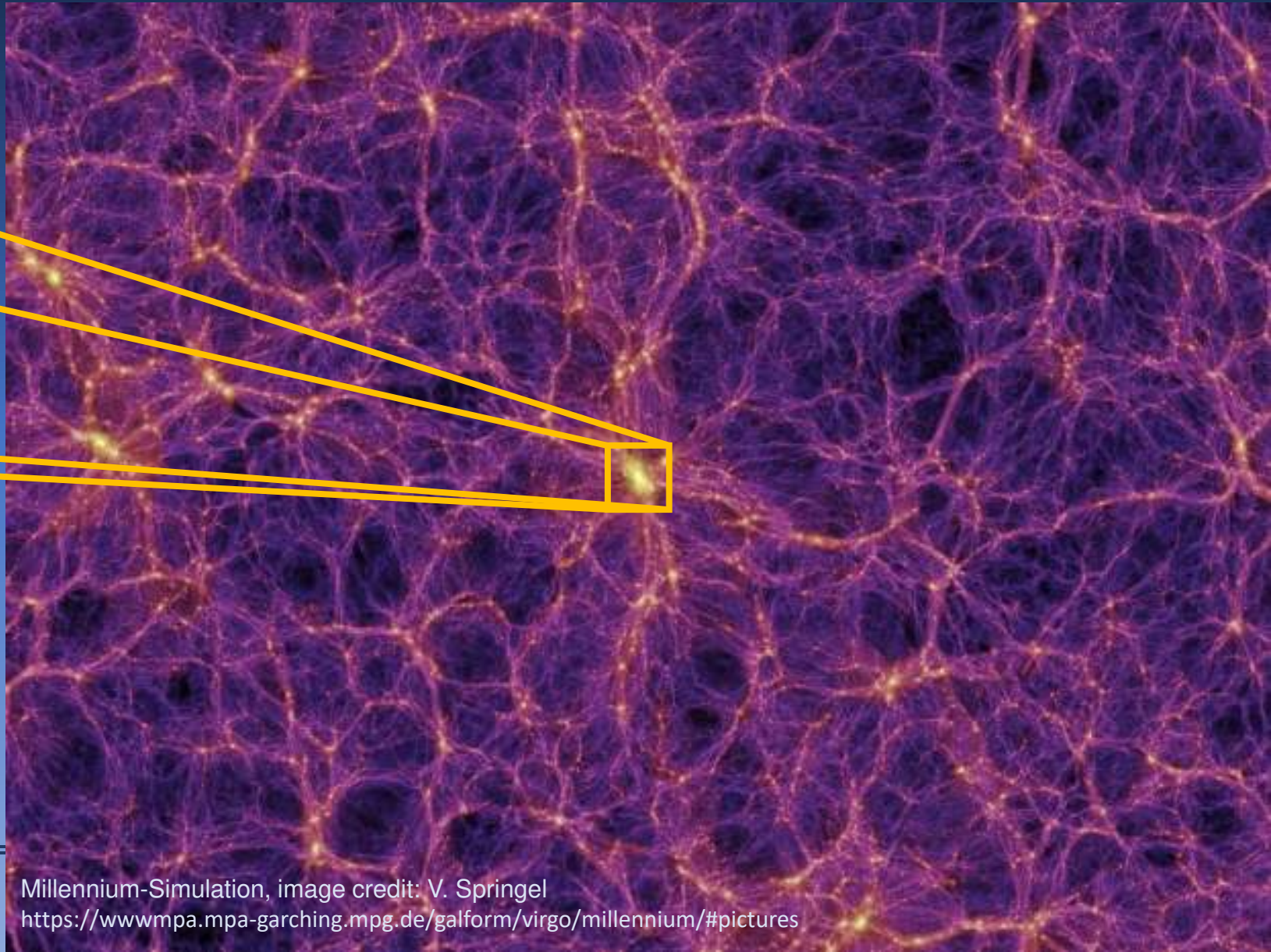
From A. Knebe: <http://popia.ft.uam.es/aknebe/page3/files/ComputationalCosmology/05ICs.pdf>

Sidestep: Semi-Analytic Models

Semi-Analytic Models and Empirical Models



Image Credit: [NASA](#), [ESA](#), Jennifer Lotz and the
HFF Team ([STScI](#))



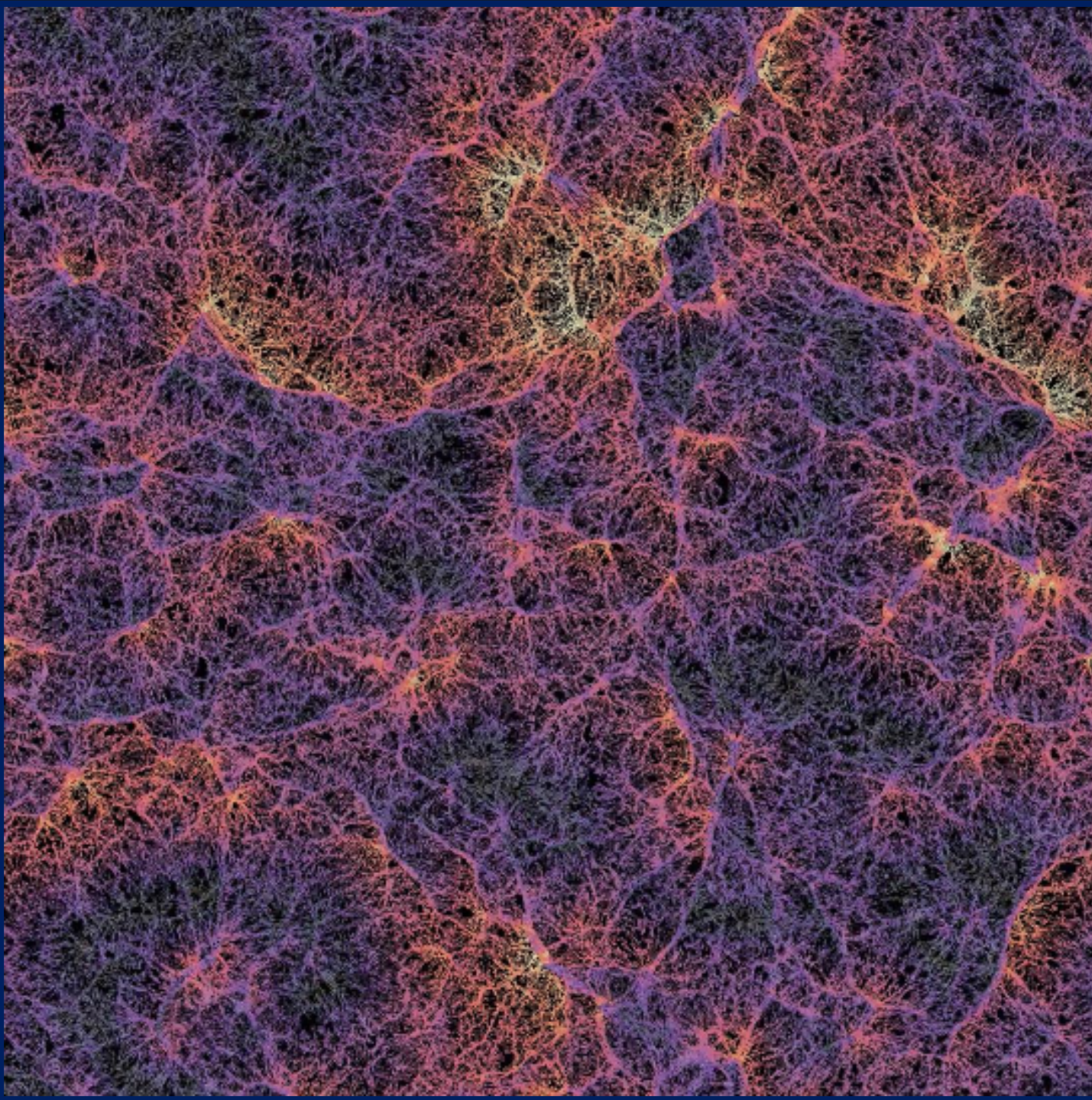
Millennium-Simulation, image credit: V. Springel
<https://wwwmpa.mpa-garching.mpg.de/galform/virgo/millennium/#pictures>

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Magneticum
Pathfinder
Simulations by Klaus
Dolag

www.magneticum.org



Magneticum Pathfinder
Simulations by Klaus Dolag

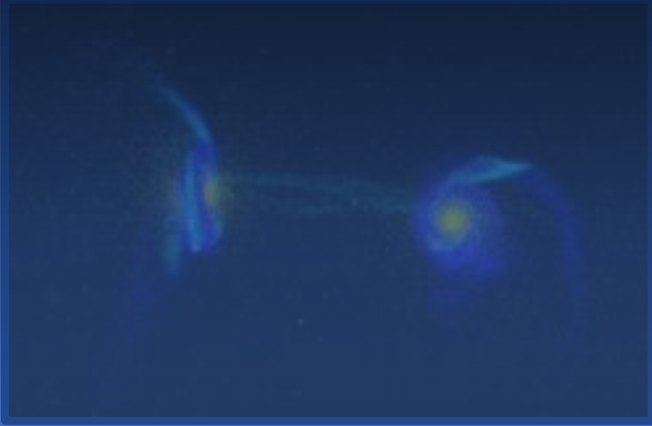
www.magneticum.org

Credit: Benjamin Seidel

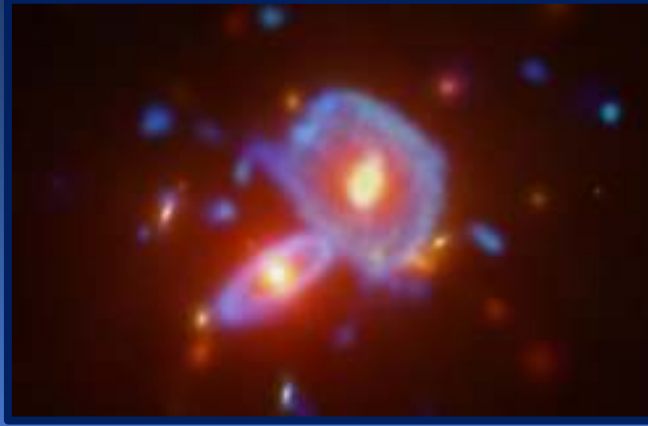


Overview: Galaxy Formation Simulations

Isolated Merger Simulations



Cosmological Zoom-Simulations



Cosmological Box Simulations

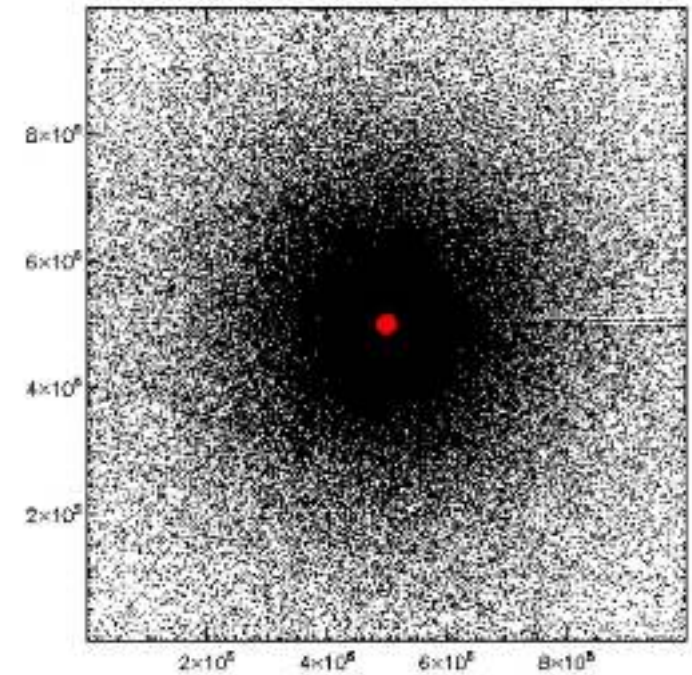
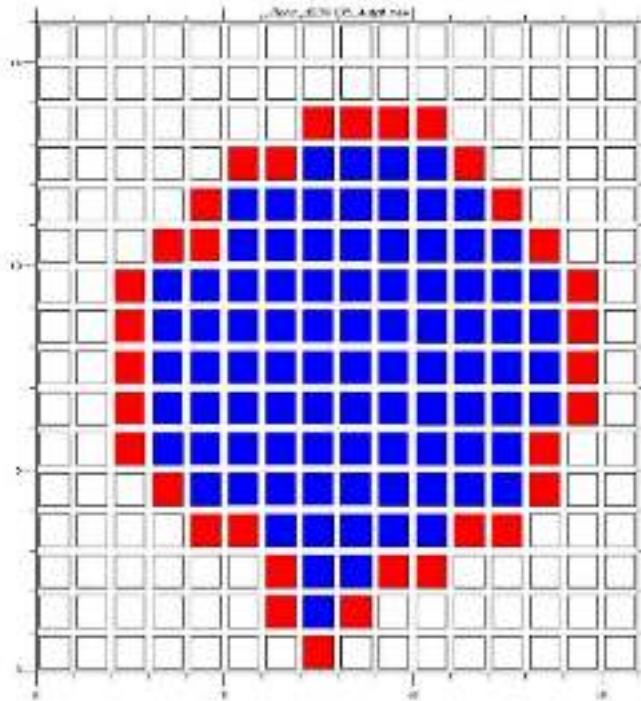


Cosmological Zoom Simulations

Initial Conditions:

Chose a galaxy of interest from a cosmological dark matter box volume.

Then split the dark matter particles into dark matter and gas in the region of interest, while the remaining box volume stays on low resolution dark matter only



From D. Schlachtberger, Master's Thesis

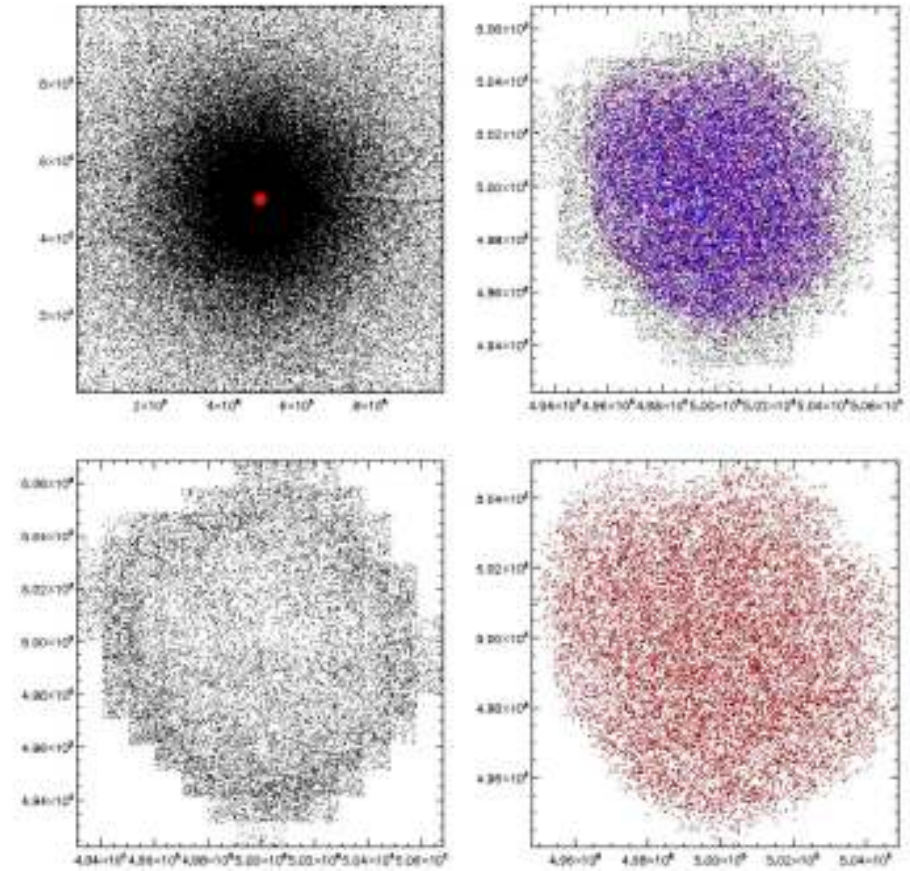
Cosmological Zoom Simulations

Initial Conditions:

Chose a galaxy of interest from a cosmological dark matter box volume.

Then split the dark matter particles into dark matter and gas in the region of interest, while the remaining box volume stays on low resolution dark matter only.

Only a small part of the cosmological box is simulated in high resolution.



From D. Schlachtberger, Master's Thesis

Cosmological Zoom Simulations



Gyr = 8.29
 $z = 0.592$

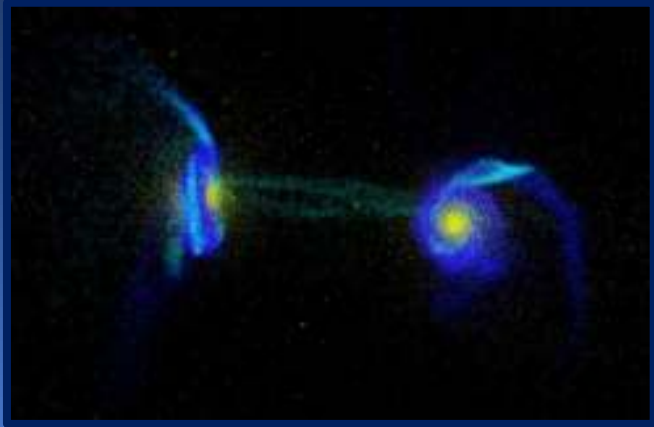


COMPASS ZOOM SIMULATIONS. Image Credit: D. Schlachtberger



Overview: Galaxy Formation Simulations

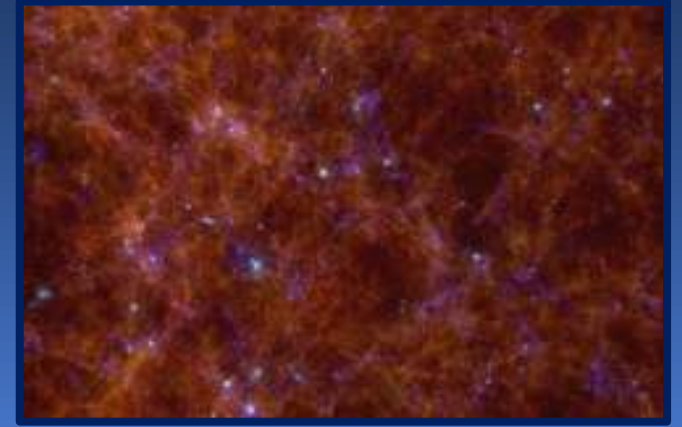
Isolated Merger Simulations



Cosmological Zoom-Simulations



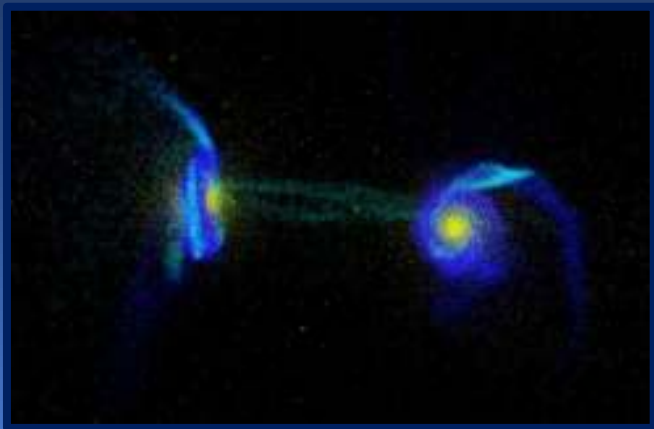
Cosmological Box Simulations





Overview: Galaxy Formation Simulations

Isolated Merger Simulations



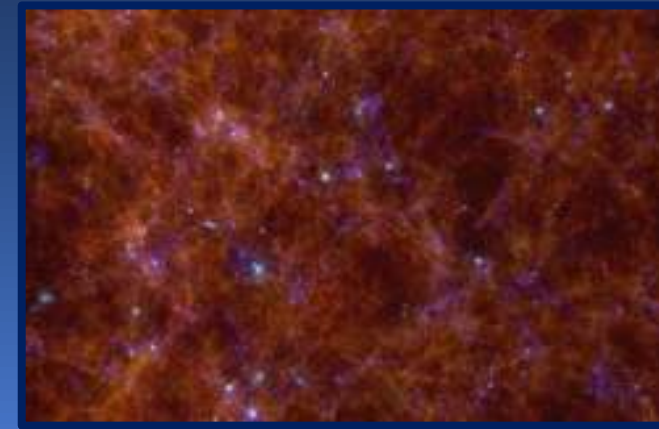
- + high resolution (temporal & spatial)
- + controlled conditions
- + parameter study (orbits, masses...)
- no cosmological formation
- not statistical representative
- no redshift evolution

Cosmological Zoom-Simulations



- + high resolution (temporal & spatial)
- + cosmological formation
- + target objects (special environment...)
- biased selection of targets
- not statistical representative
- no parameter control

Cosmological Box Simulations

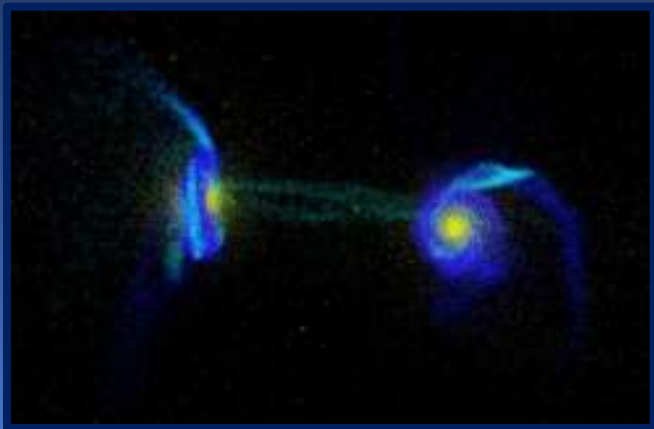


- + cosmological formation
- + statistical representative
- + no selection bias
- low resolution
- no parameter control
- computationally expensive



Overview: Galaxy Formation Simulations

Isolated Merger Simulations



There are a lot of different idealized merger simulations, but most of them do not have a fancy name

Cosmological Zoom-Simulations



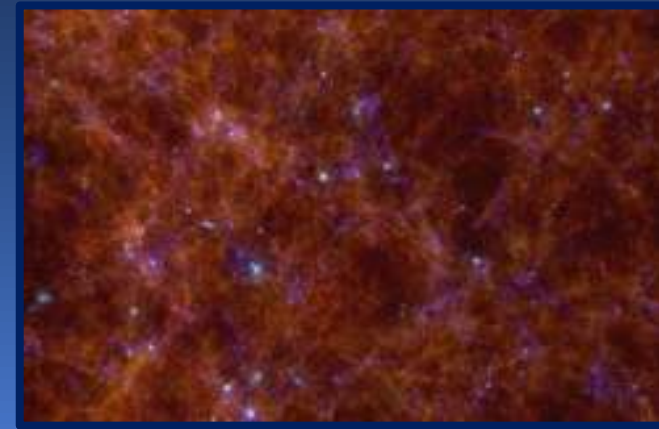
Galaxies (Dwarf to Groups)

- Oser/Naab
- NiHAO
- Fire
- Aquarius
- E-Mosaics

Clusters:

- The 300 Project
- CEagle/Hydrangea
- Romulus-C

Cosmological Box Simulations



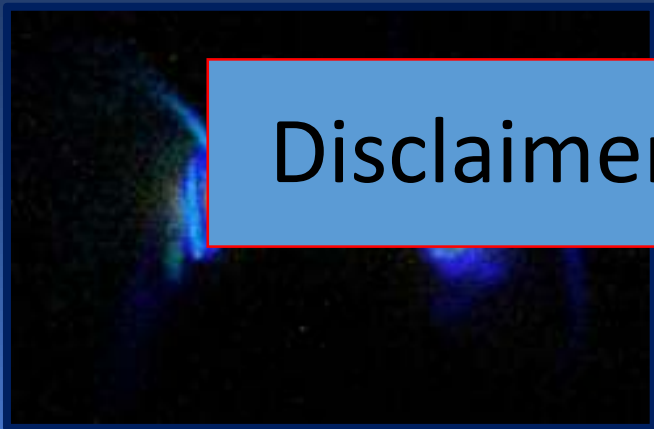
Fully Hydrodynamical

- Eagle
- Magneticum
- HorizonAGN
- Illustris/IllustrisTNG
- MassiveBlack



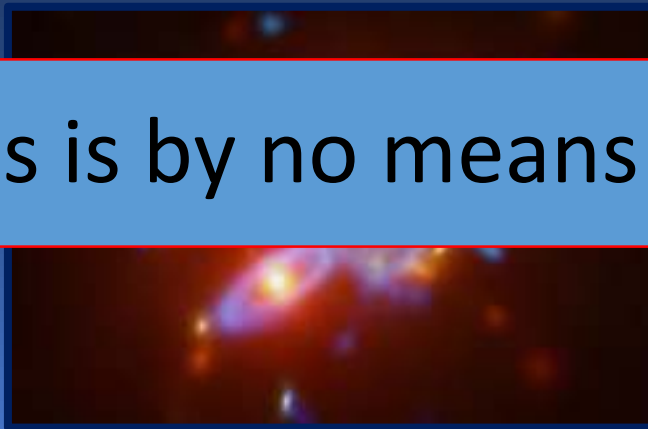
Overview: Galaxy Formation Simulations

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There are a lot of different idealized merger simulations, but most of them do not have a fancy name

Cosmological Zoom-Simulations



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Cosmological Box Simulations



Fully Hydrodynamical

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Disclaimer: This is by no means a complete list!



Overview: Galaxy Formation Simulations

Isolated Merger Simulations



There are a lot of different idealized merger simulations, but most of them do not have a fancy name

Cosmological Zoom-Simulations



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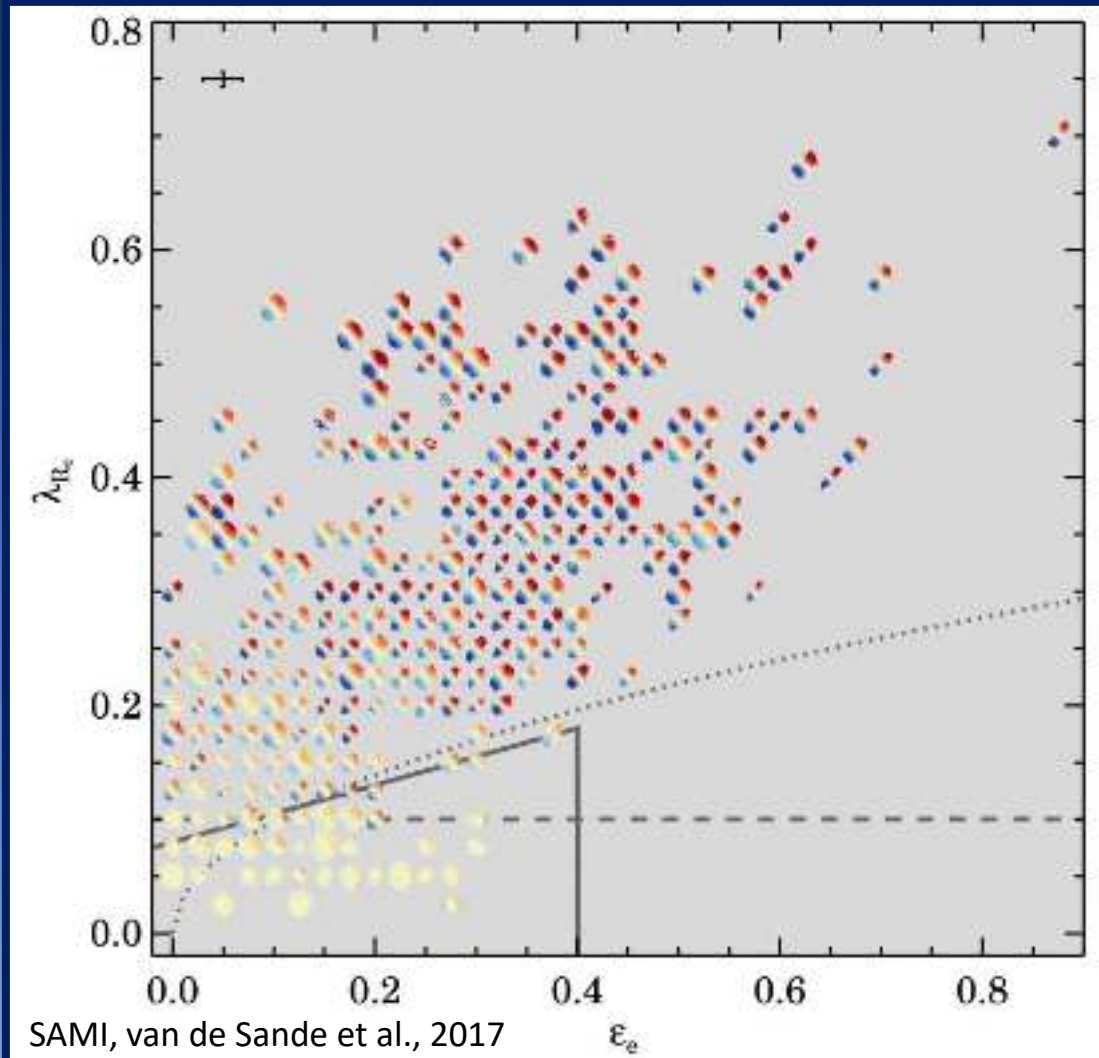
Fully Hydrodynamical

- Eagle
- Magneticum
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- Illustris/IllustrisTNG
- MassiveBlack

Why do we need them all?

Why so many different simulation types?

Example: Why different Cosmological Simulations

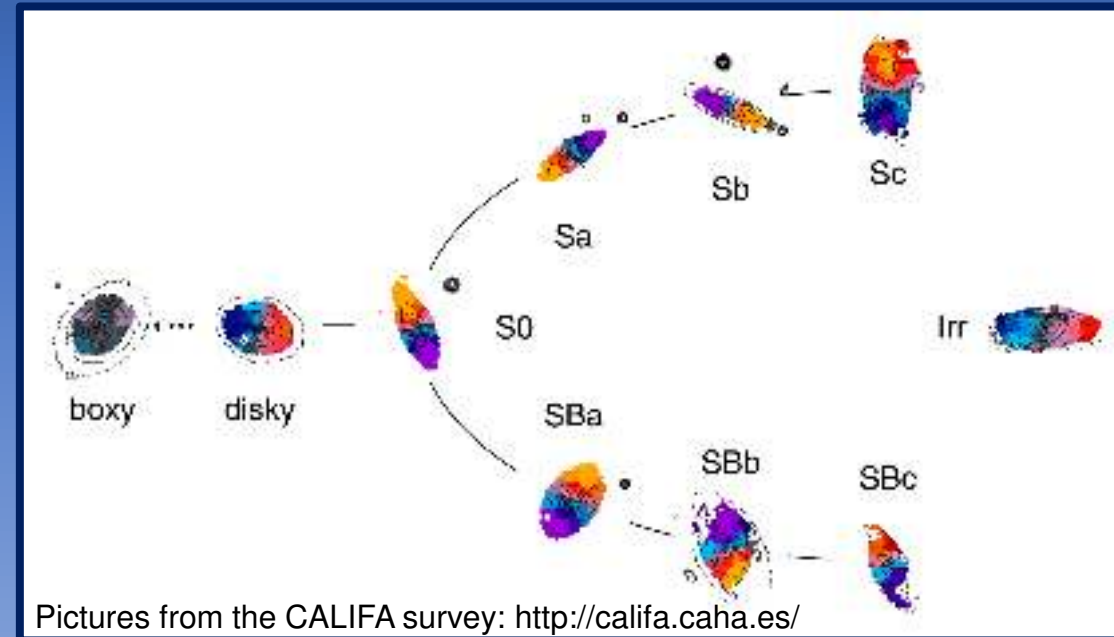


Fast
rotating
ETGs

Slow
rotating
ETGs

$$\lambda_R \equiv \frac{\langle R | V | \rangle}{\langle R \sqrt{V^2 + \sigma^2} \rangle}$$

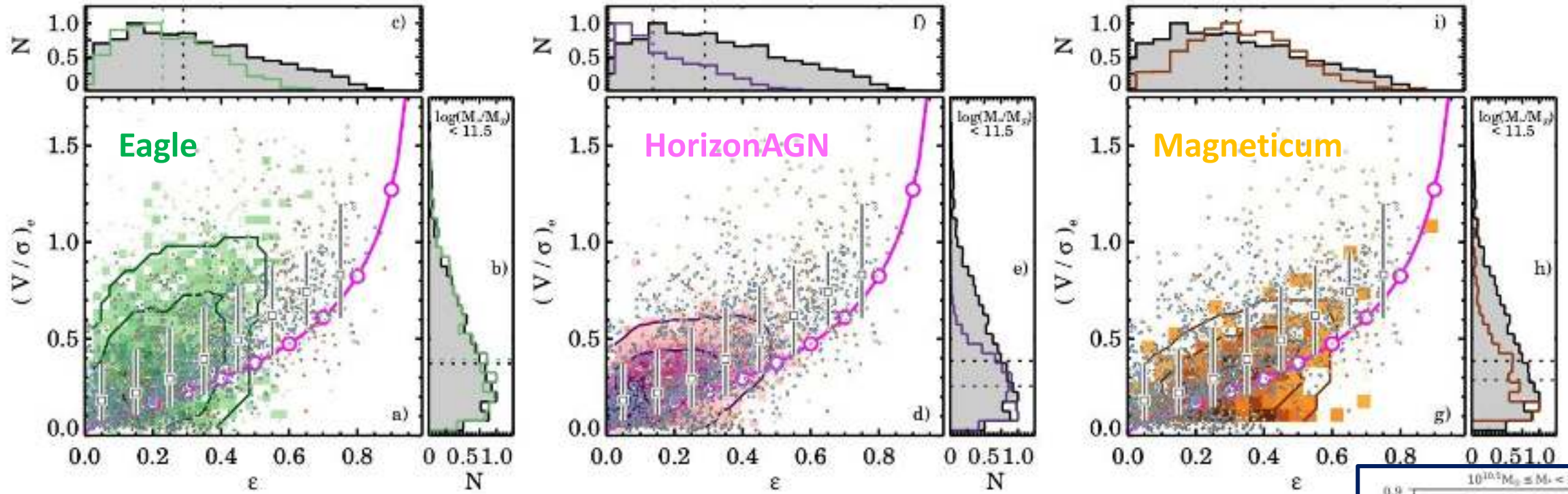
Emsellem et al., 2007; 2011



Pictures from the CALIFA survey: <http://califa.caha.es/>

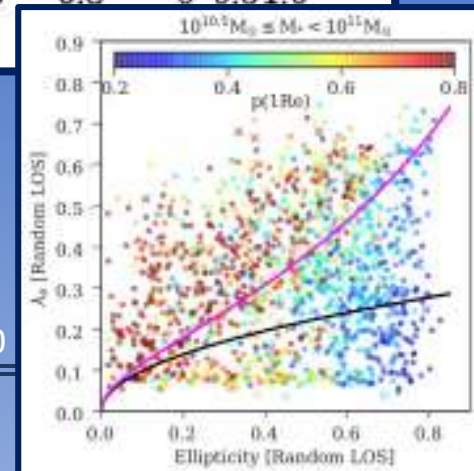


Simulations versus Observations



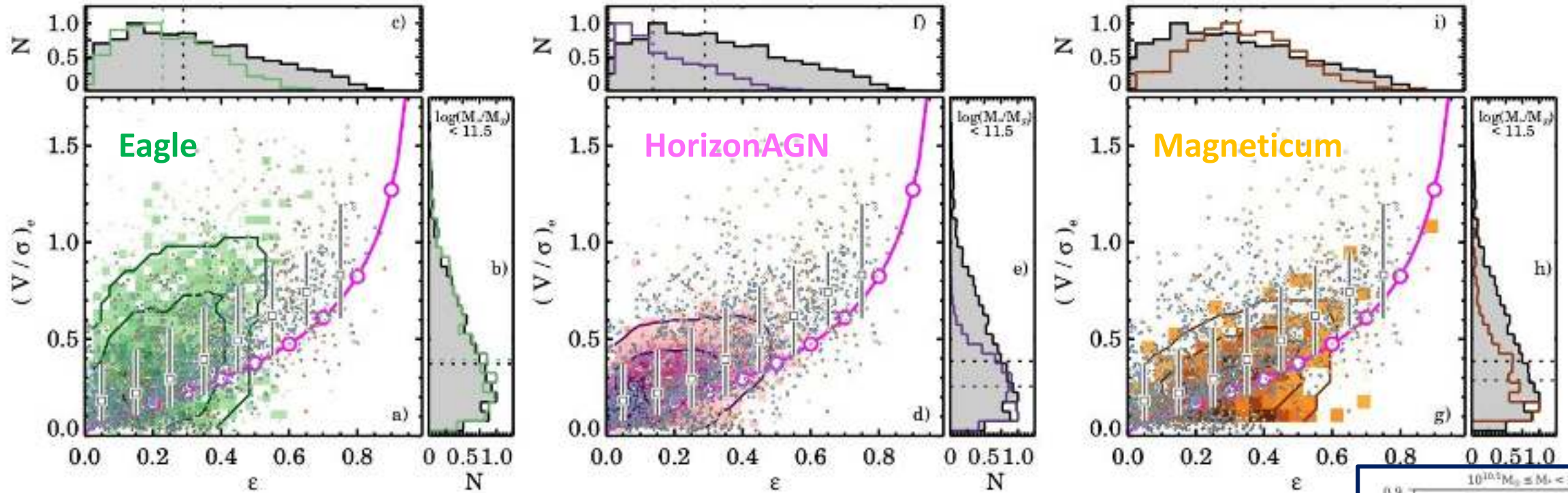
Van de Sande et al., 2019

TNG: Pulsoni et al., 2020





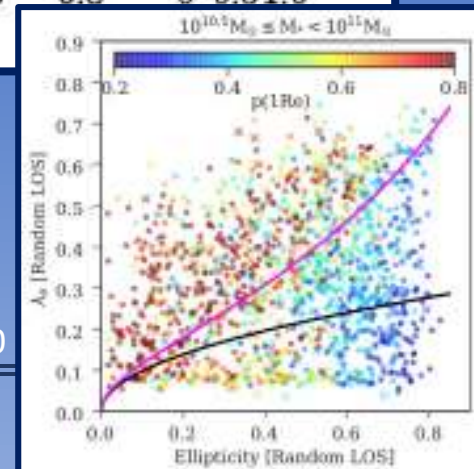
Simulations versus Observations



Van de Sande et al., 2019

Differences between cosmological simulations are not from the different Codes but from the SubGrid Physics

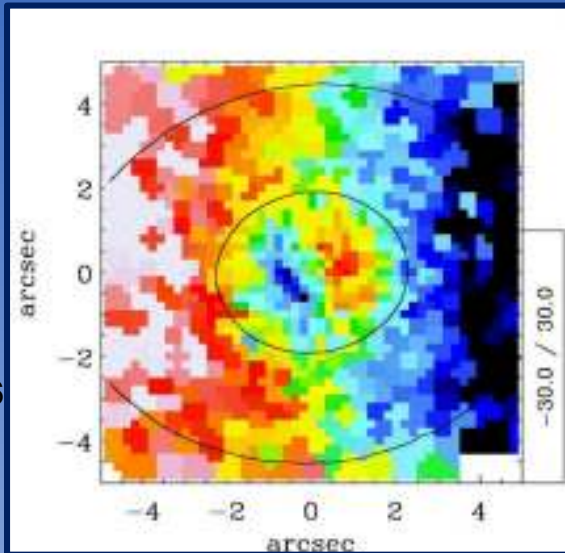
TNG: Pulsoni et al., 2020



Example: Why different Galaxy Simulations?

Kinematically Distinct Cores in ETGs come in two flavors:

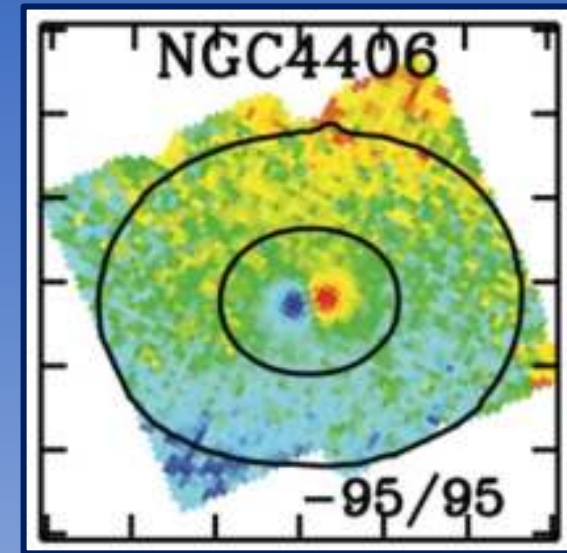
Small ($<1\text{kpc}$) core with a young stellar population, mostly inside fast rotators



McDermid et al., 2006
but also in dwarf
galaxies (Rys et al.,
2013)



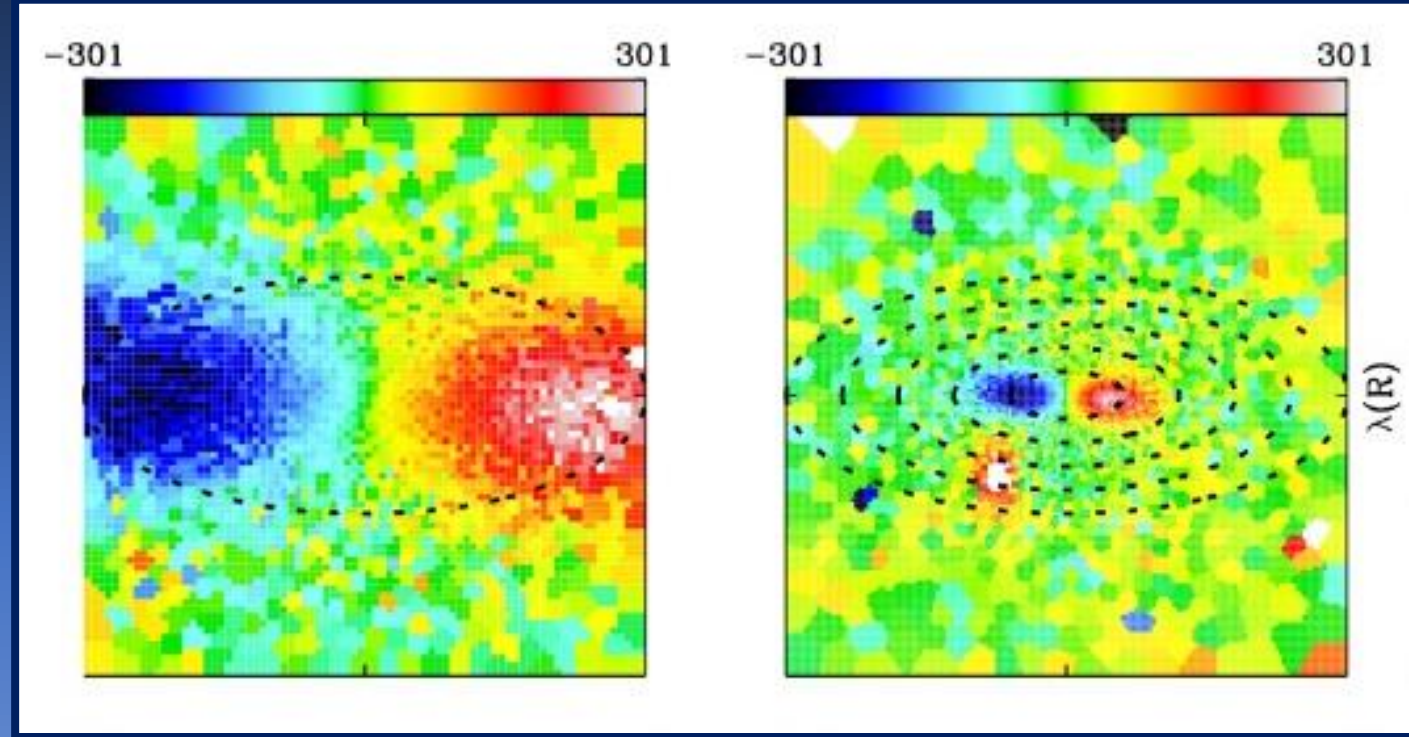
Large core with an old stellar population, usually in massive slow rotators



Kranjovic et al.,
2013

Old KDCs: Cosmological Simulation

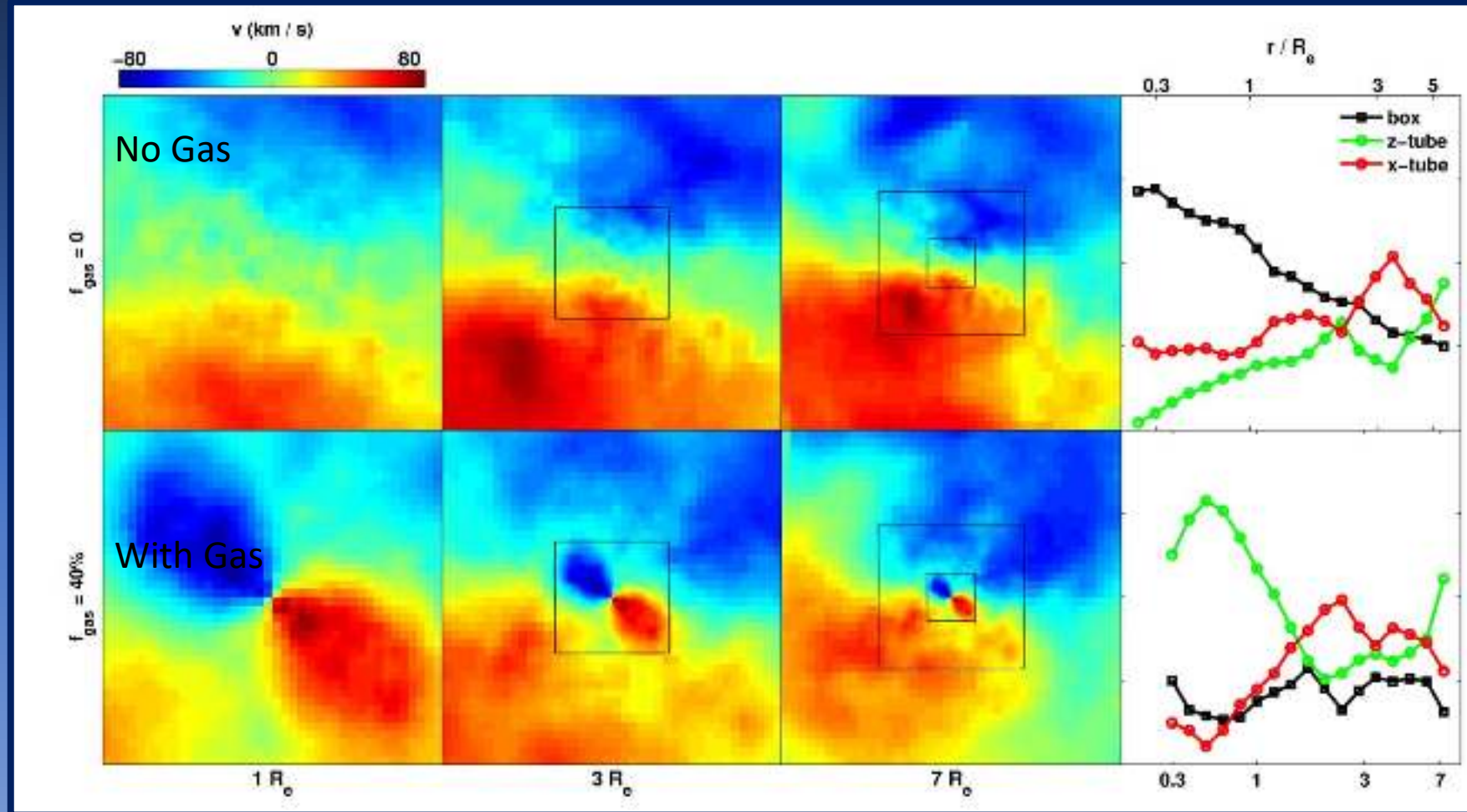
Large KDCs are found in cosmological simulations to be made from an old rotating disks already build up at $z=2$, which then only accreted minor or mini mergers with ratios smaller than 1:10. These mergers destroy the rotation in the outskirts, but cannot reach the central areas of a galaxy (see Schulze et al., 2020 and Karademir et al., 2019)



Schulze et al., 2020; see also Schulze et al., 2018 for kinematic maps within $1R_{half}$ and correlations with kinematic type

Young KDCs: Isolated Merger Simulations

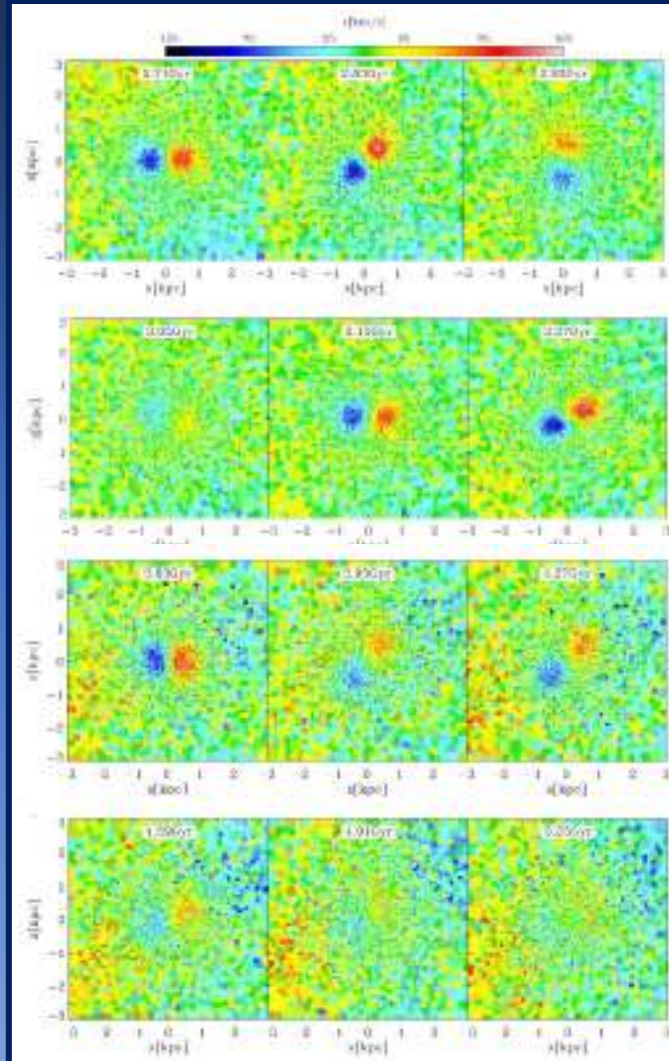
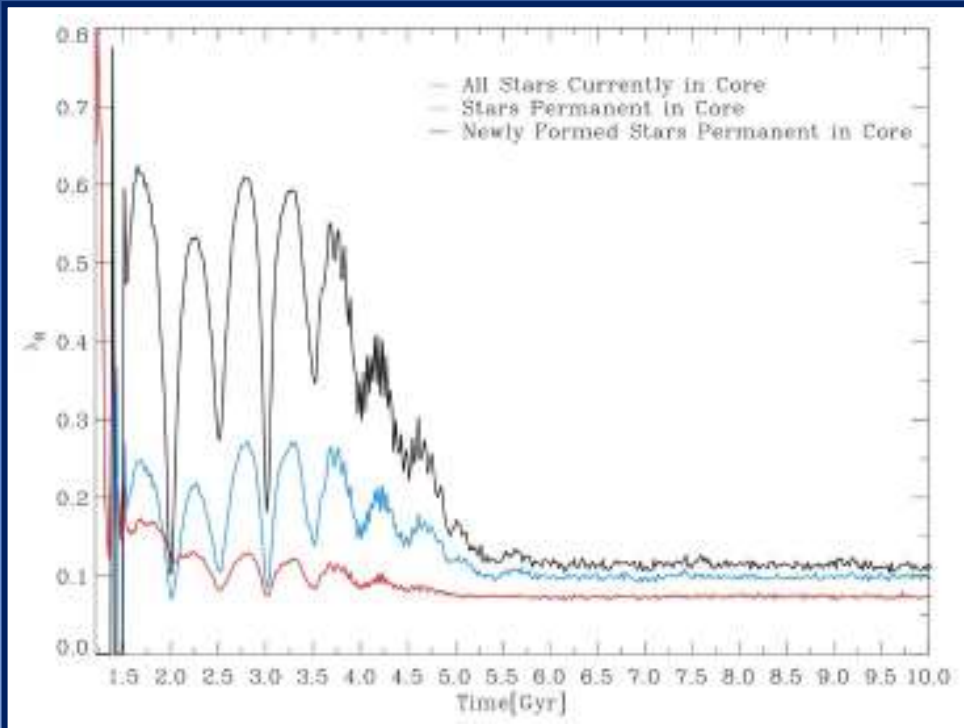
Young, small KDC are formed in gas-rich major merger events, and they are dominated by z-tube orbits. They are only formed by mergers with gas fractions of 15-40%



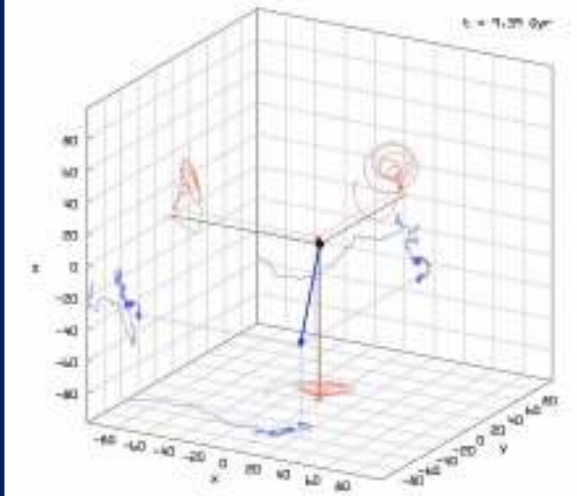
Hoffman et al., 2010

Young KDCs: Isolated Merger Simulations

KDC made from mostly newly-formed stars that keep the memory of the angular momentum of the merger: performs a motion comparable to the precession of a gyroscope in a gravitational potential; superposition of an intrinsic rotation and a global precession that gets gradually damped over cosmic time.



Movie by T. Hoffmann



Lifetime of a young KDC: stable for ≈ 3 Gyrs after the merger, damped over ≈ 1.5 Gyrs.

Schulze et al., 2017



Chose the right simulation type for your question!

Comparison to Observations

I. Abell 2744: Substructures in Clusters

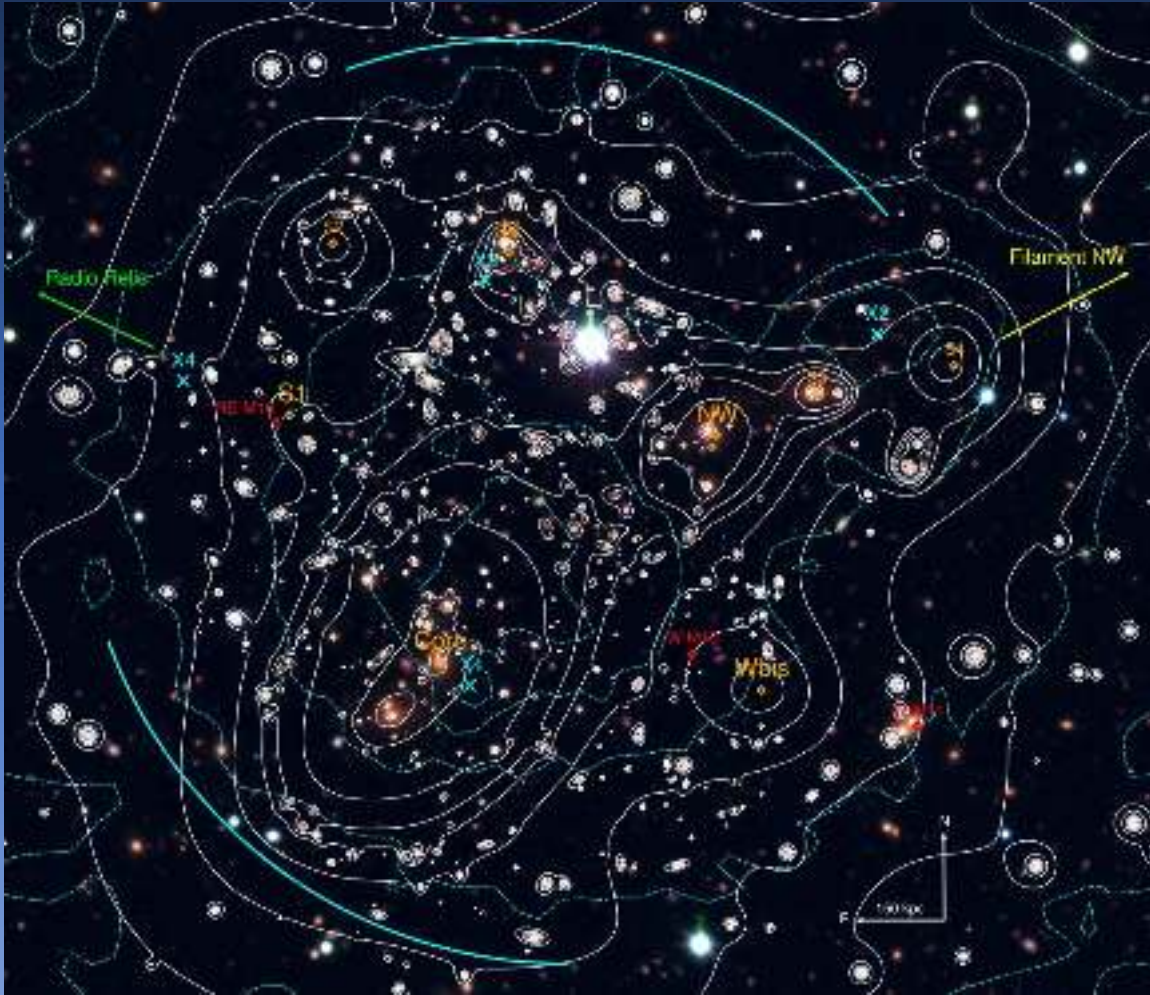


Image credit: Jauzac et al. 2016

Abell 2744

- $z = 0.308$
- Dynamically active
- In 1.3Mpc aperture:

$$M_{\text{tot}}(< 1.3\text{Mpc}) = 2.3 \cdot 10^{15} M_{\odot}$$

I. Abell 2744: Substructures in Clusters



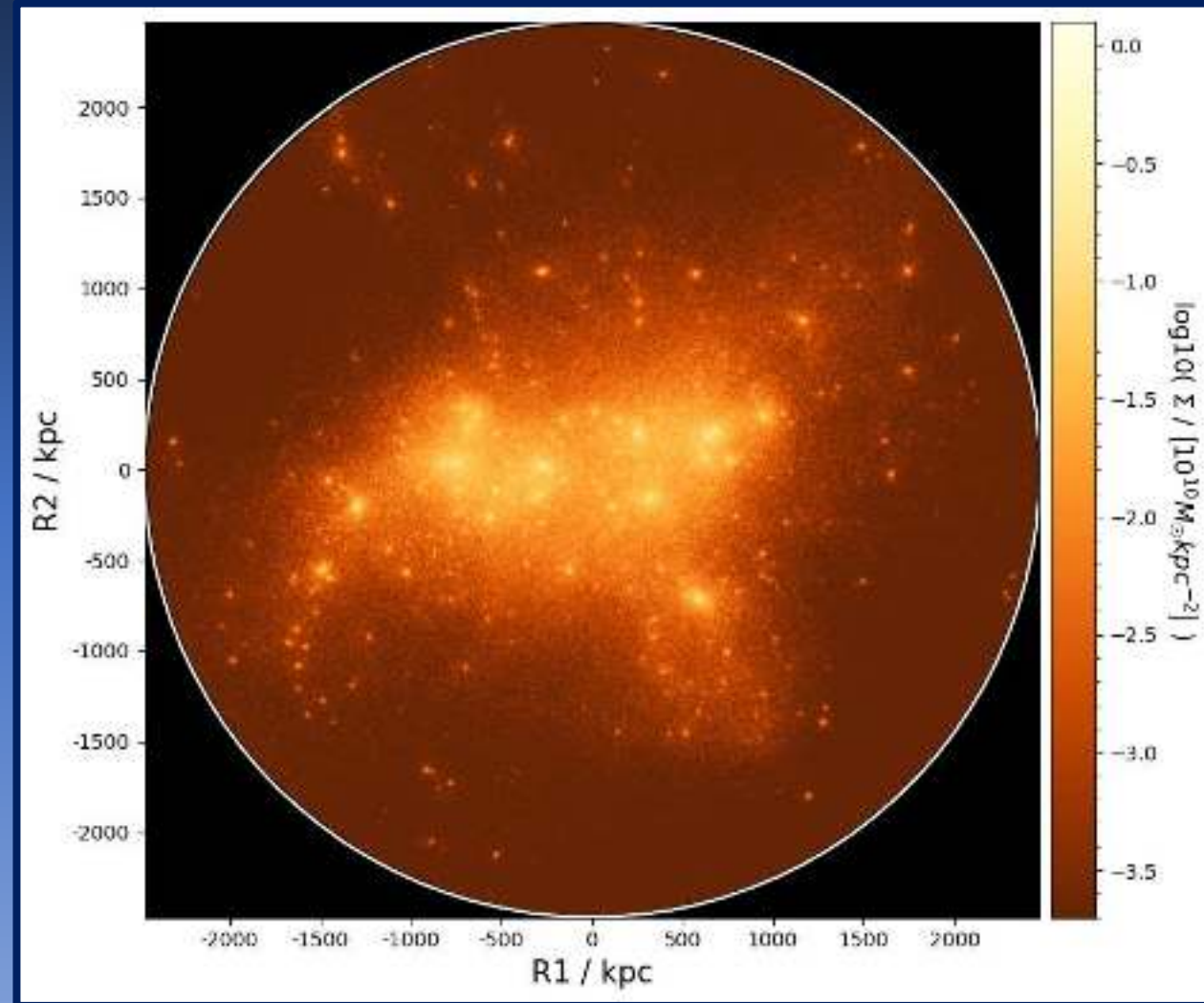
Image credit: Jauzac et al. 2016

Abell 2744

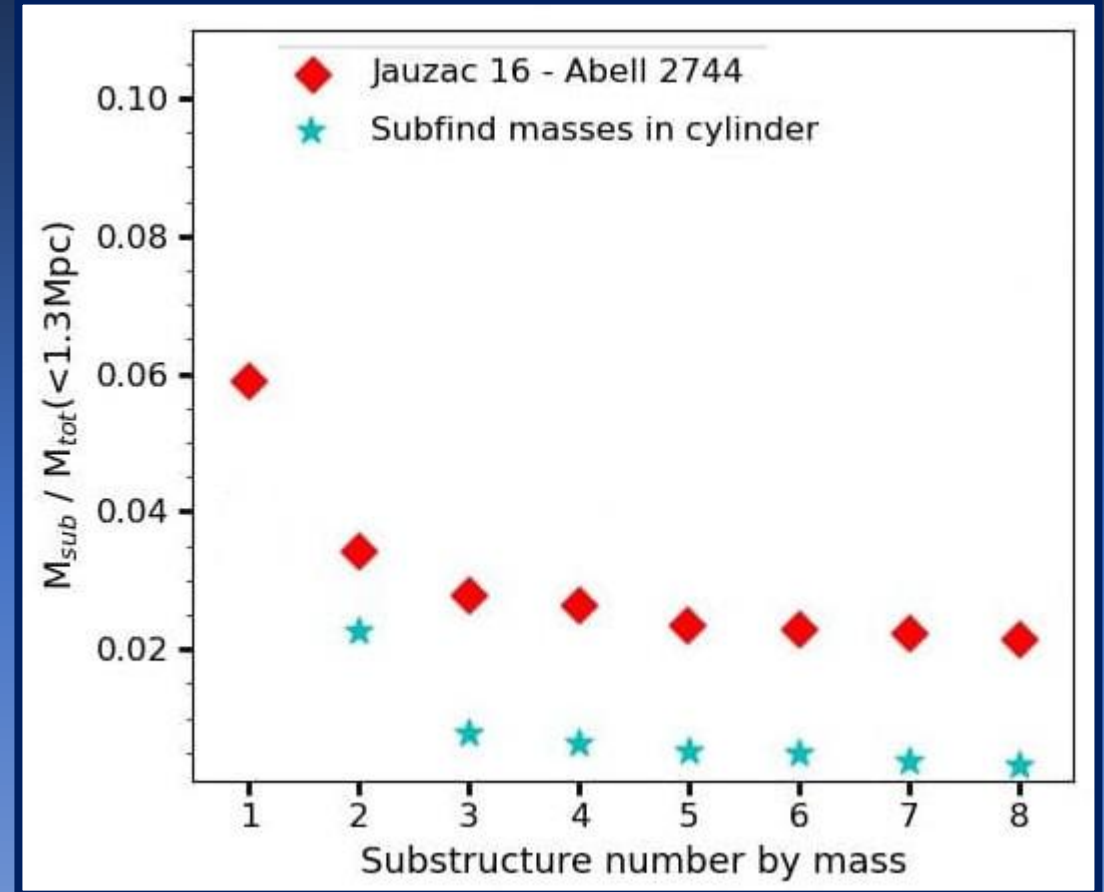
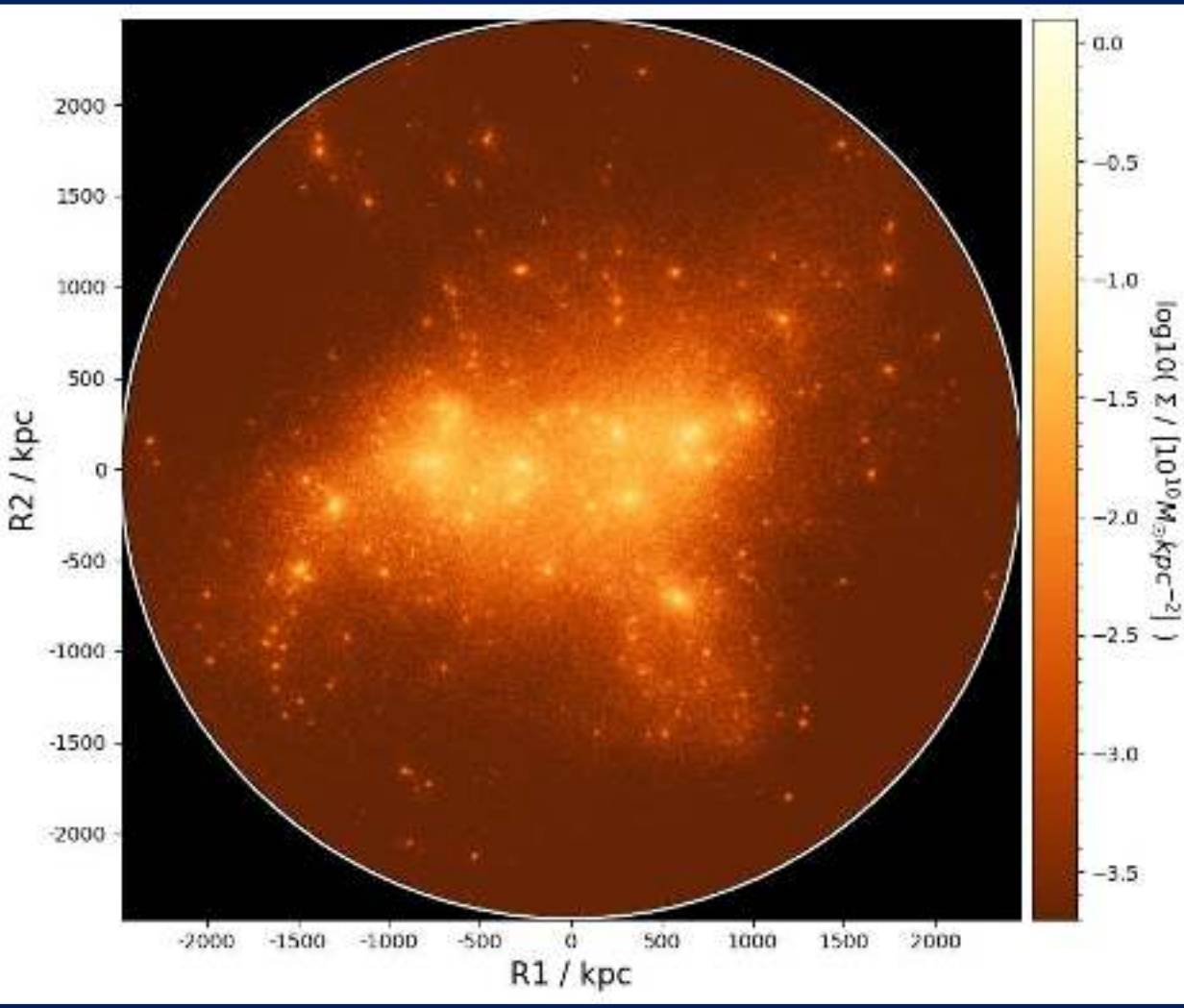
- $z = 0.308$
- Dynamically active
- In 1.3Mpc aperture:

$$M_{\text{tot}}(< 1.3\text{Mpc}) = 2.3 \cdot 10^{15} M_{\odot}$$

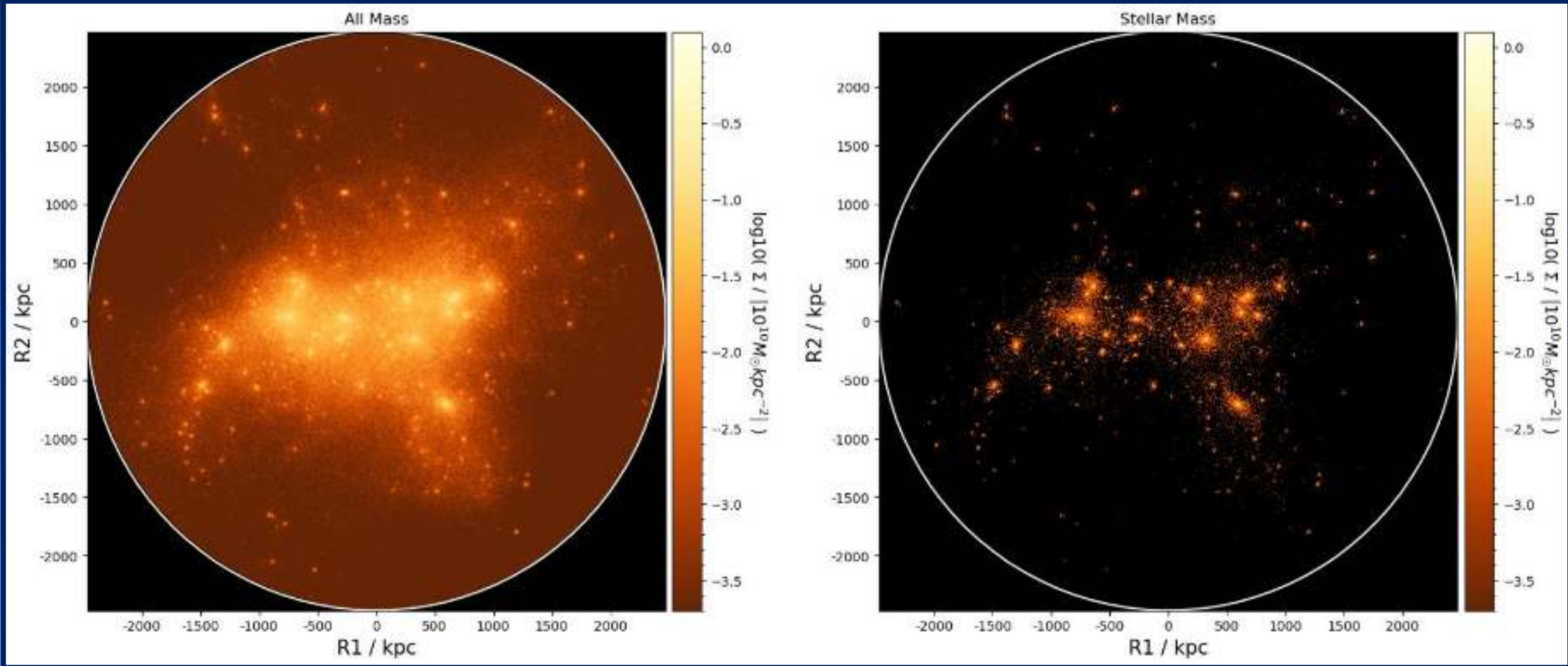
I. Abell 2744: Substructures in Clusters



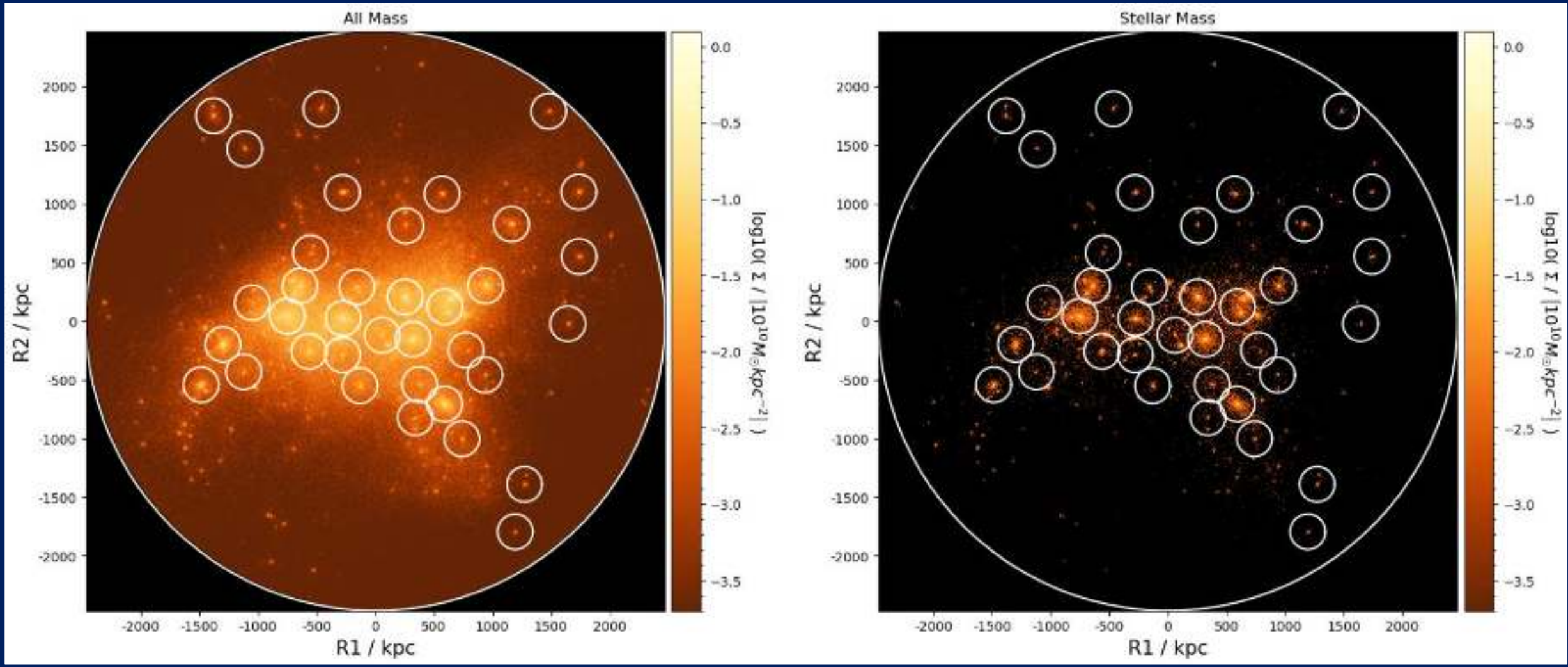
I. Abell 2744: Substructures in Clusters



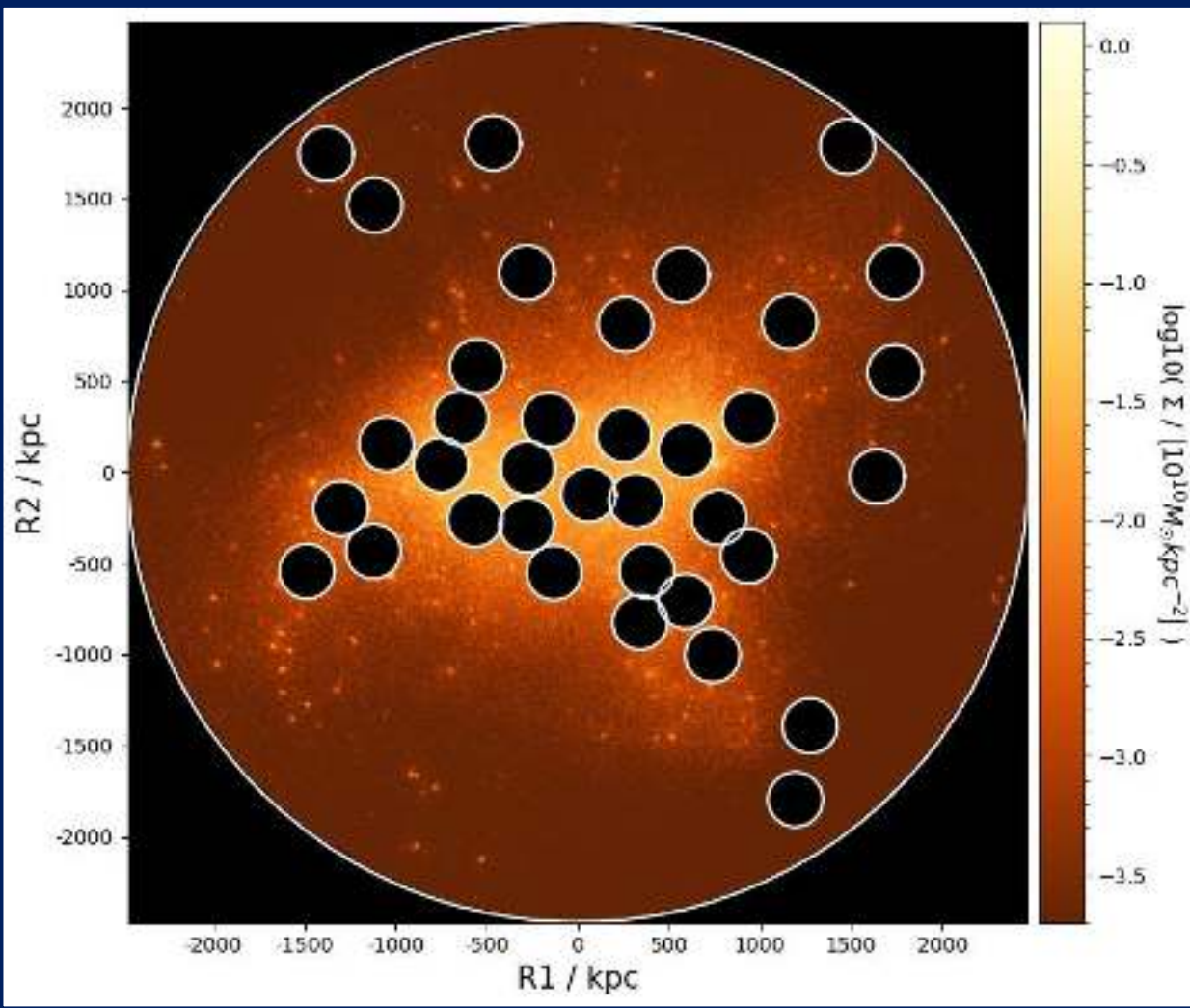
I. Abell 2744: Substructures in Clusters



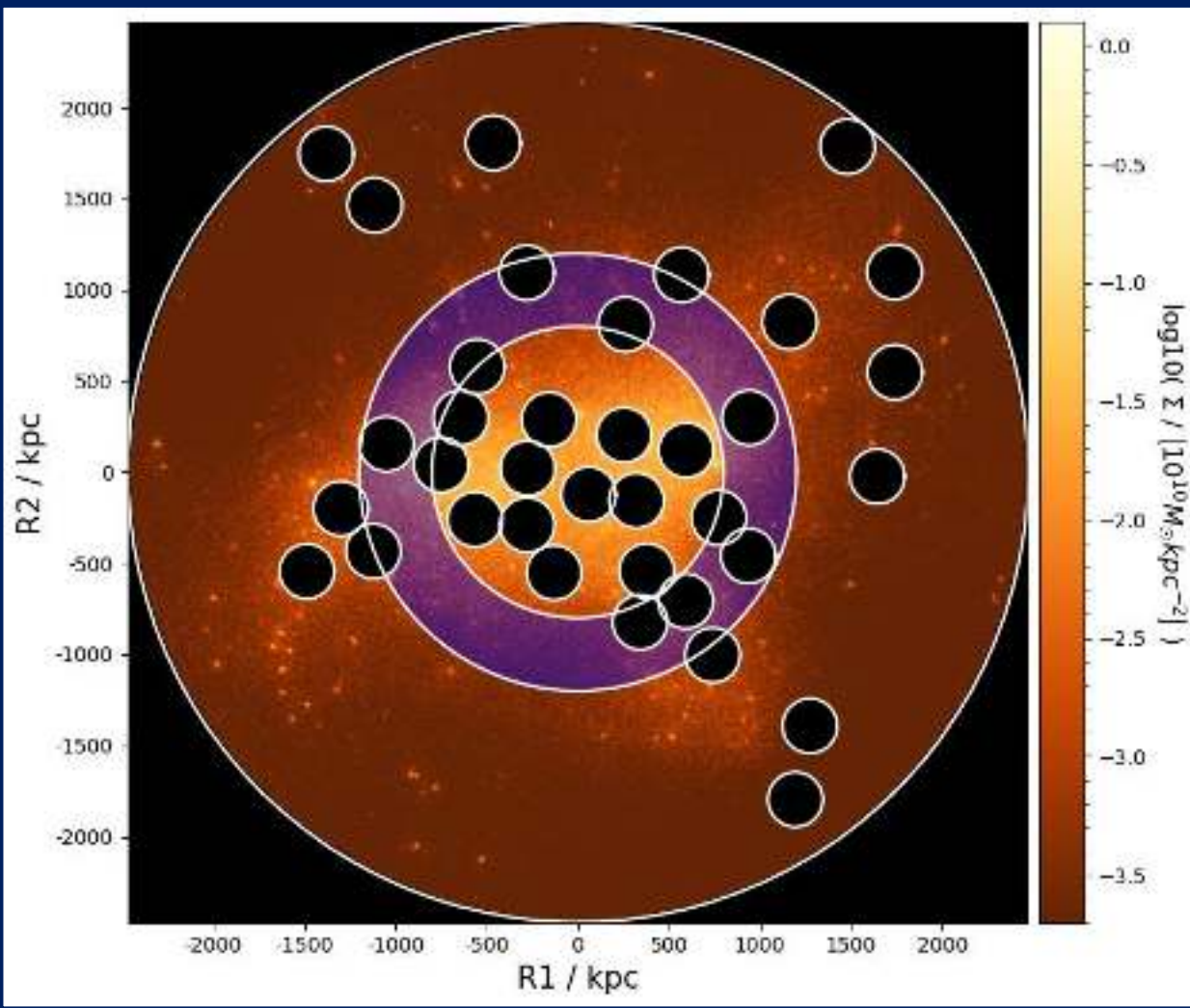
I. Abell 2744: Substructures in Clusters



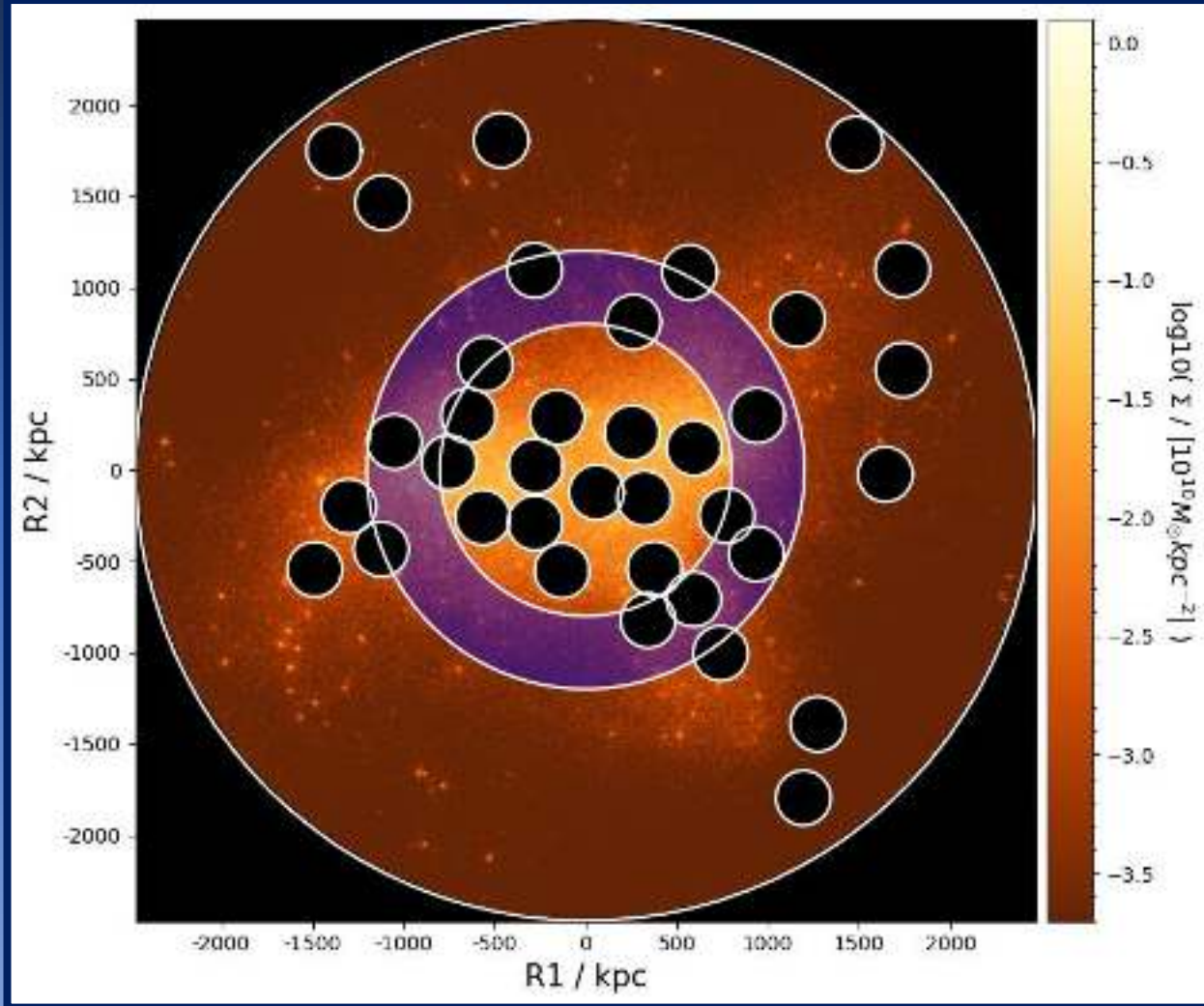
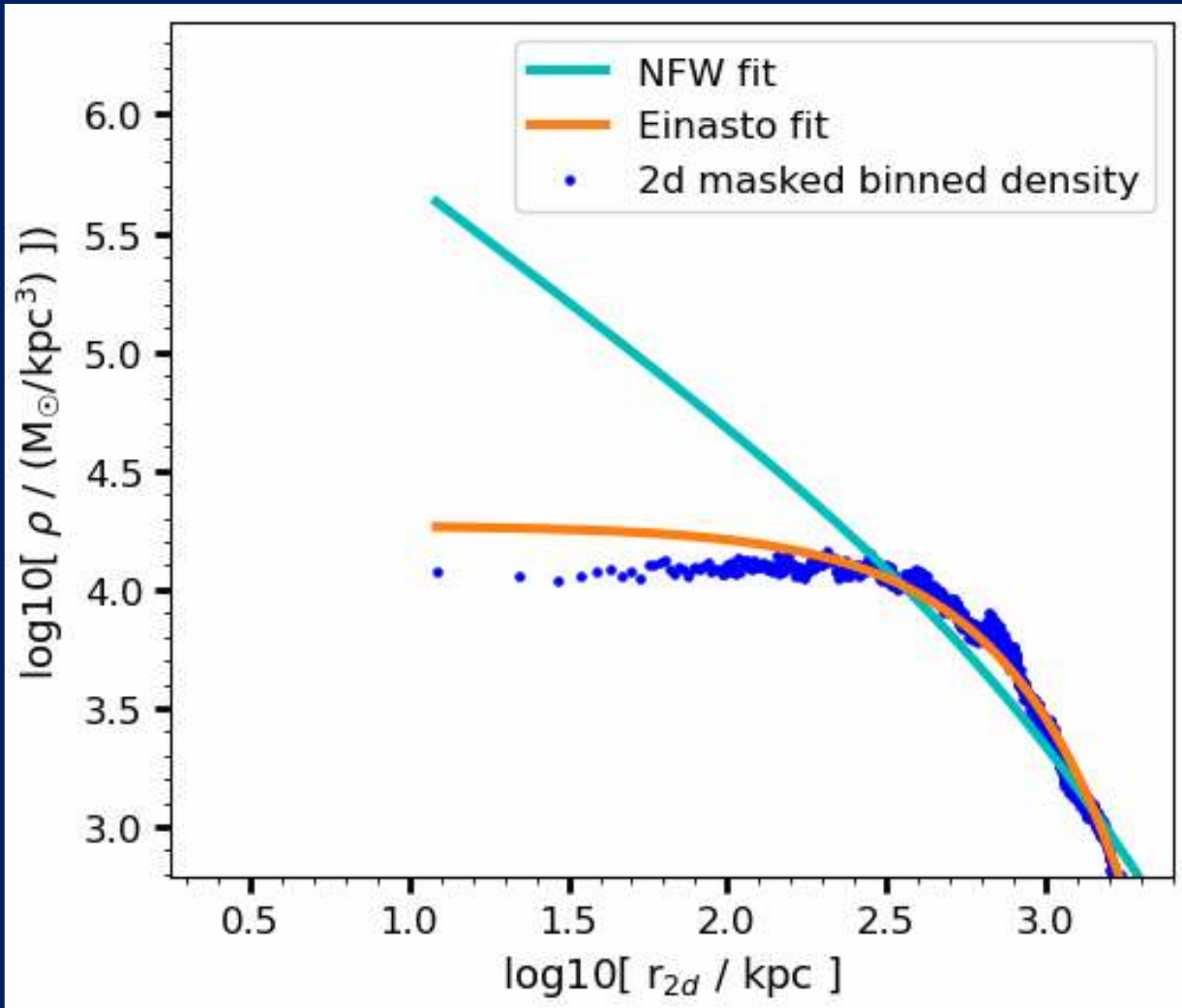
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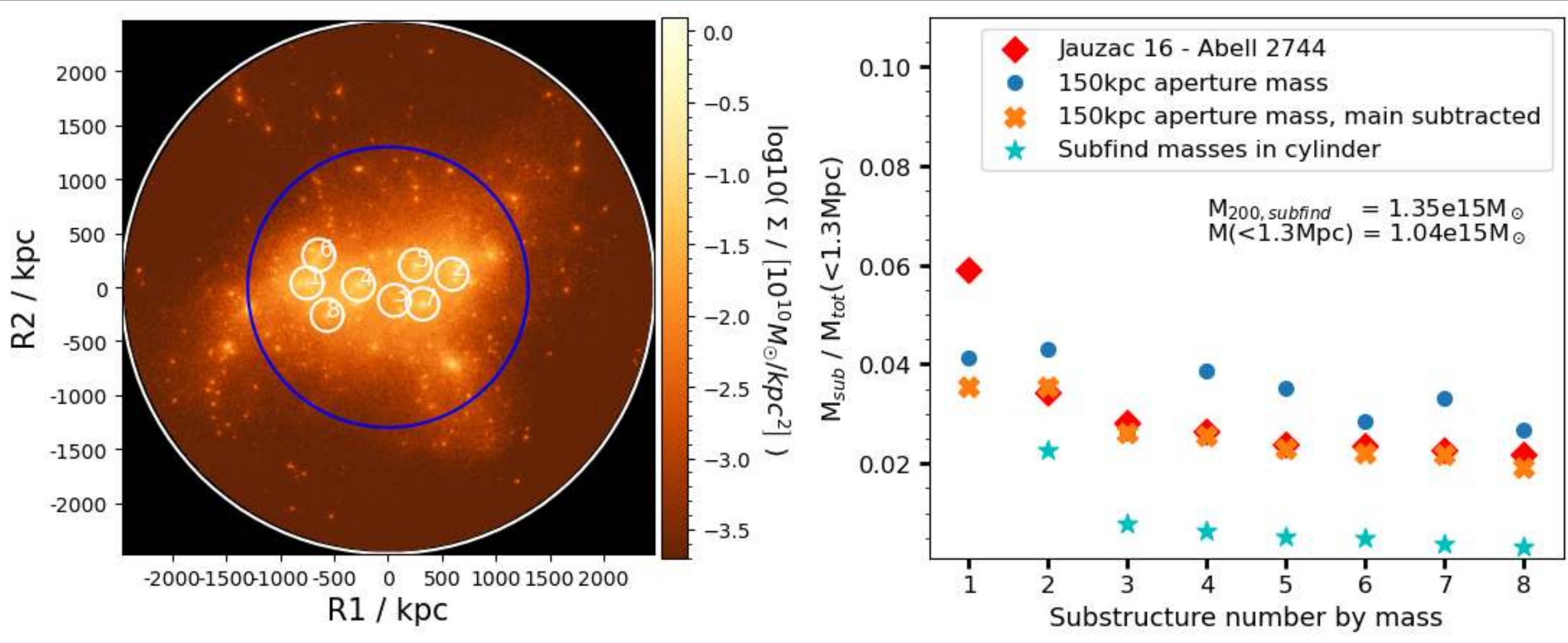
I. Abell 2744: Substructures in Clusters



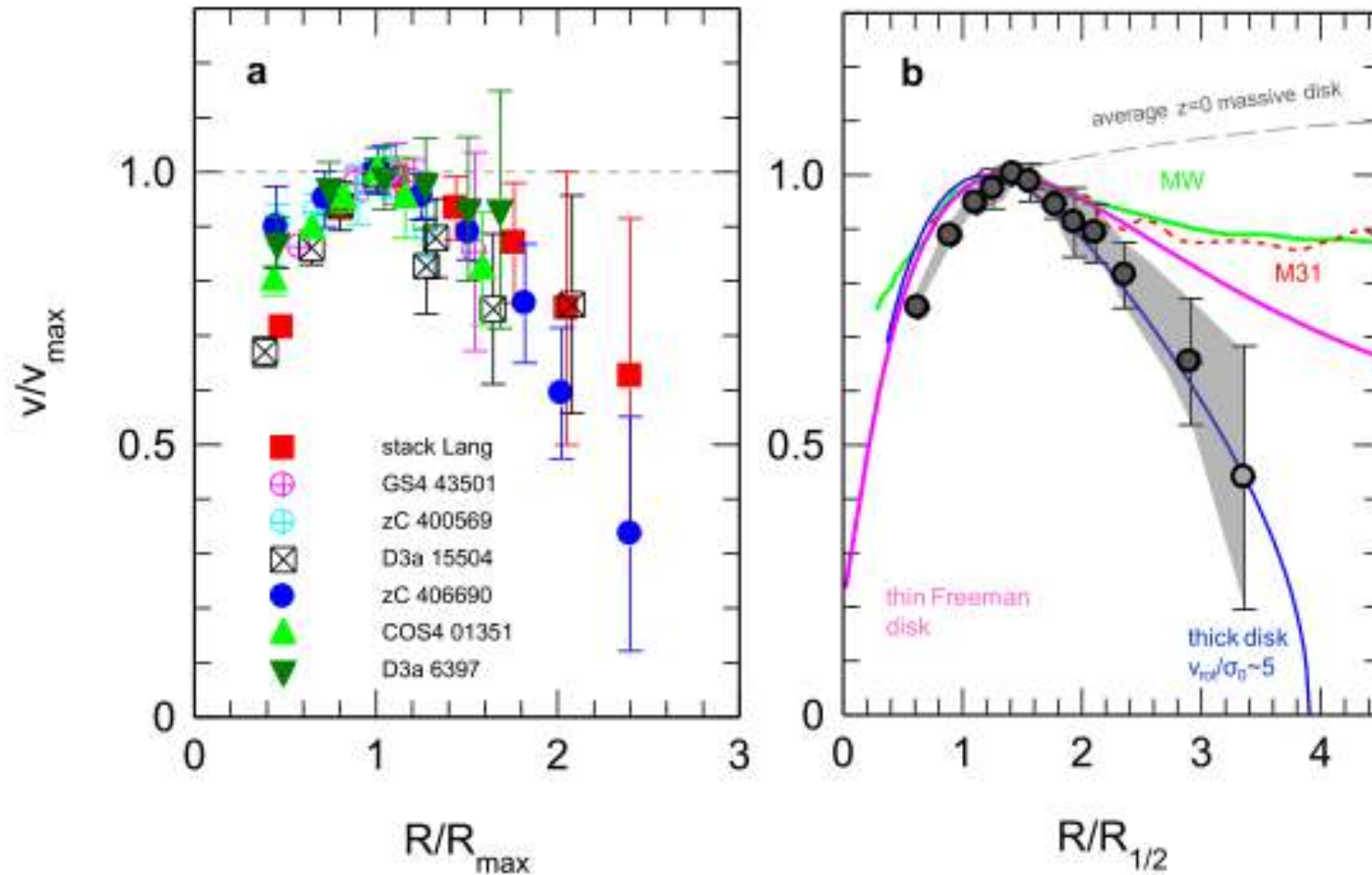
I. Abell 2744: Substructures in Clusters



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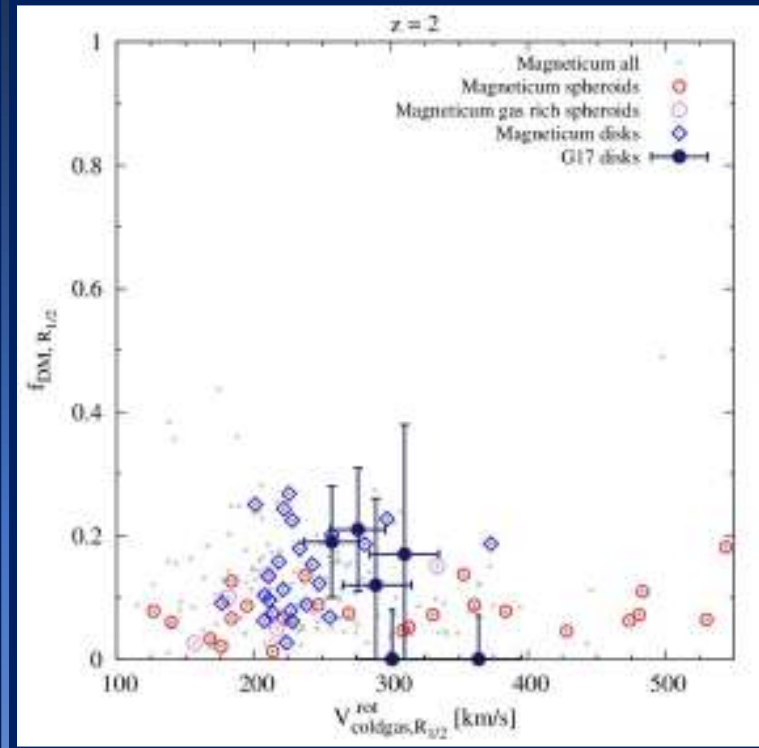
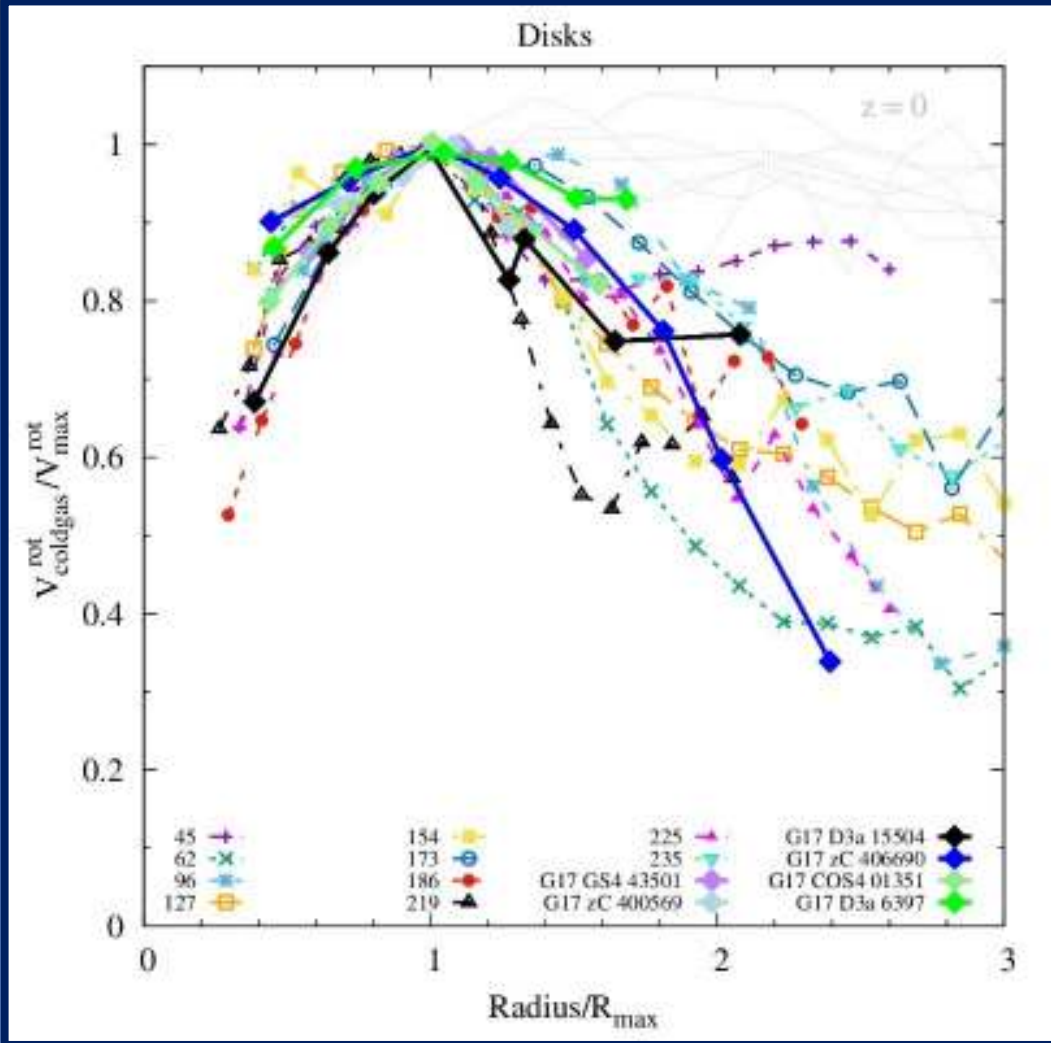


II. No Dark Matter in Disks at $z=2$?



Genzel et al., 2017

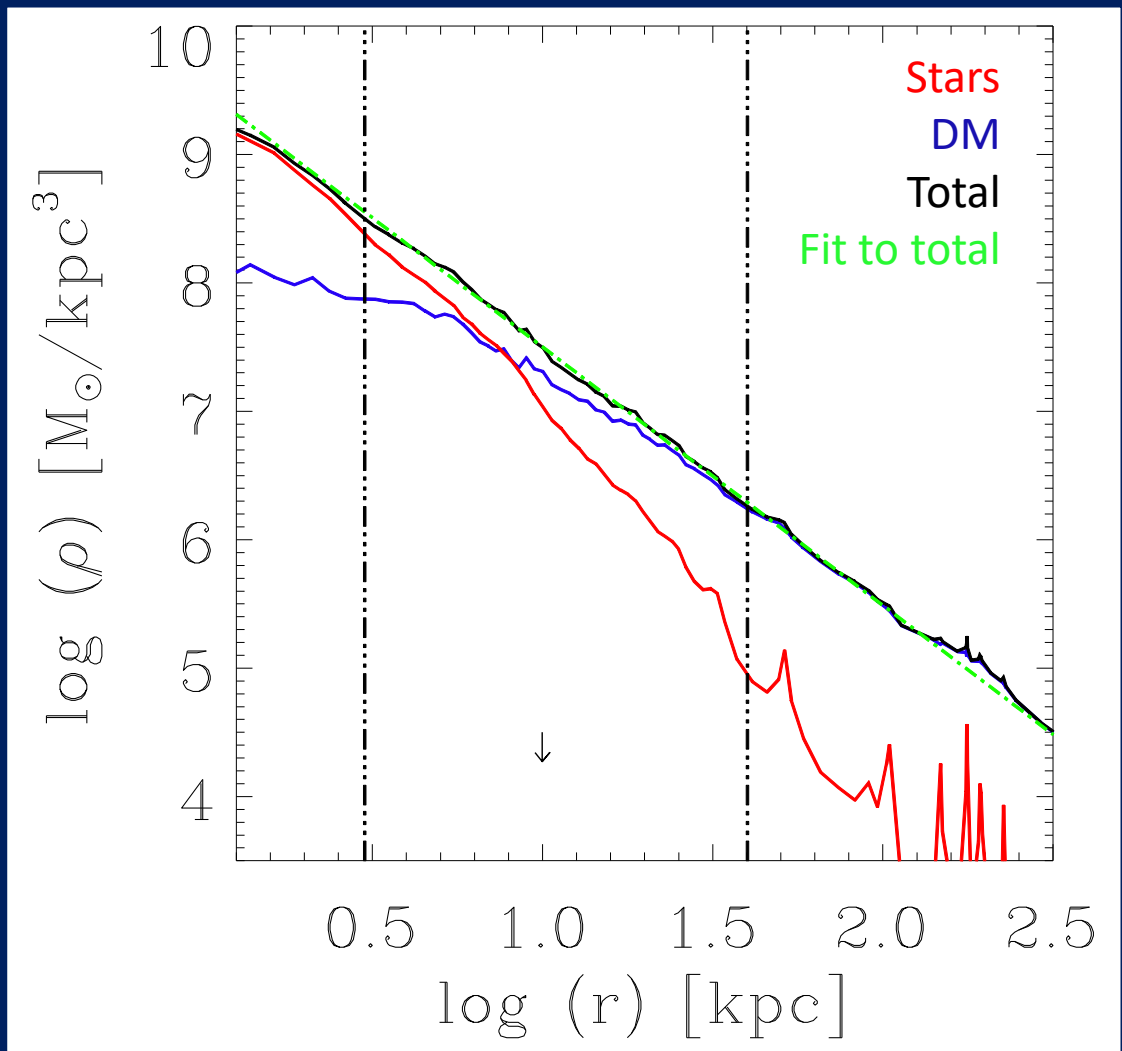
II. No Dark Matter in Disks at $z=2$?



At $z=2$, approximately half of the Magneticum disk galaxies have declining rotation curves in agreement with observations by Genzel et al., 2017, and similar central dark matter fractions.

Teklu et al., 2018

III. Total Density Slopes with Redshift



Total radial density profiles can be fit by a single power law.

Inner part: Stars dominate the total profiles.

Outer part: Dark Matter dominates the total profiles.

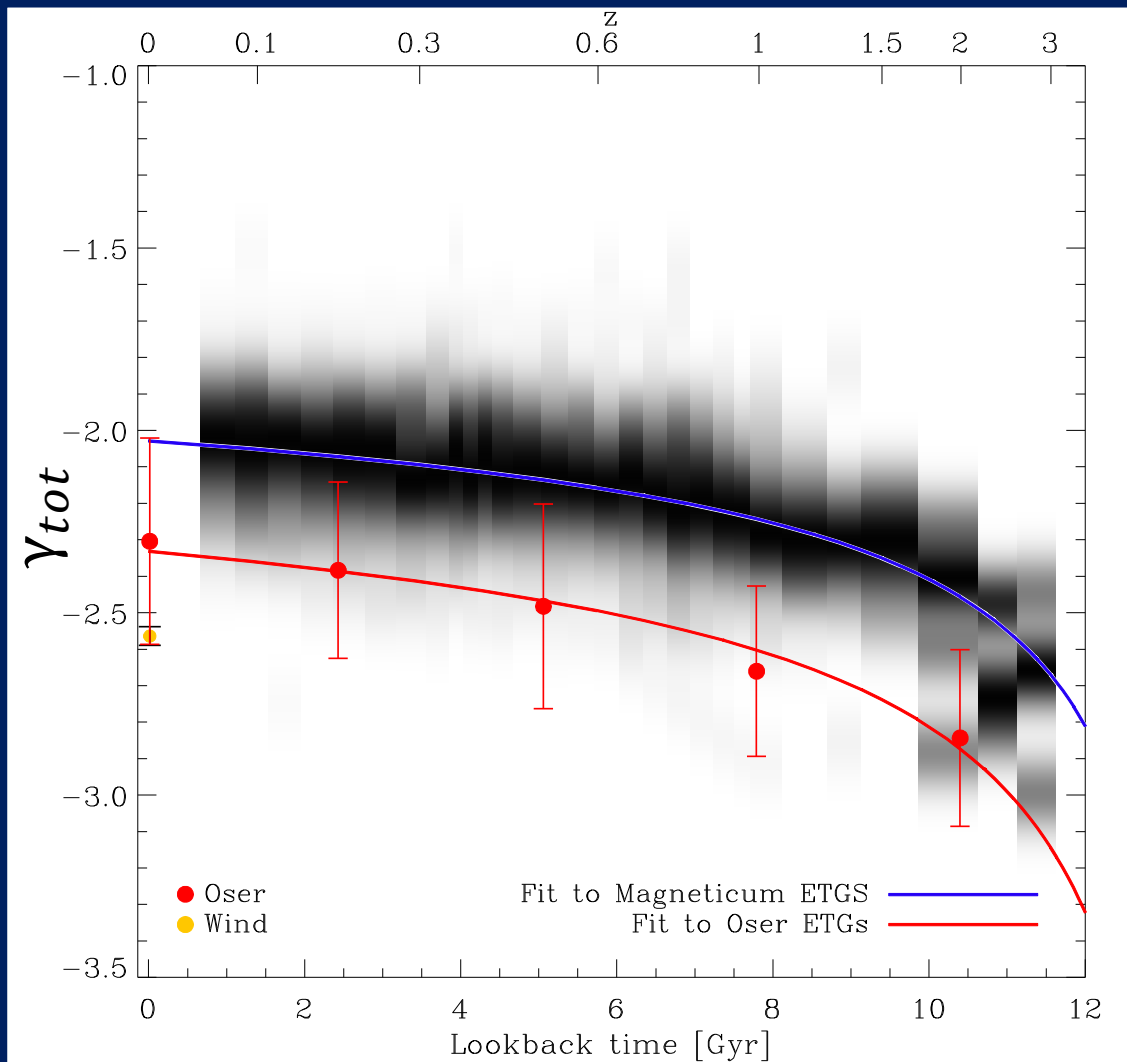
$$\gamma = \frac{d \log(\rho)}{d \log(r)}$$

Most ETGs have slopes close to isothermal, i.e. $\gamma_{\text{tot}} \approx -2$, but they can be as steep as $\gamma_{\text{tot}} \approx -3$.



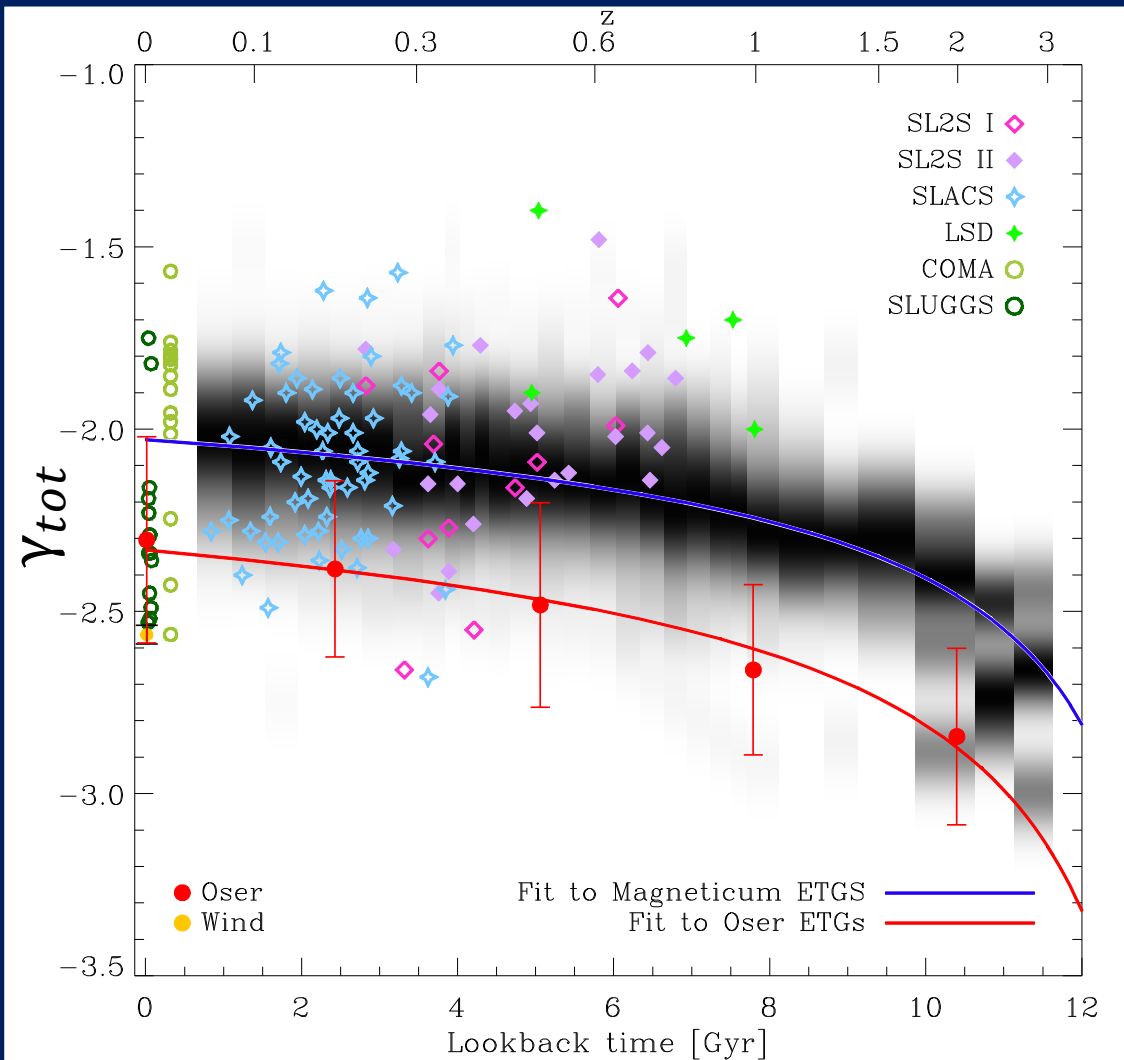
This is independent of the included feedback models

III. Total Density Slopes with Redshift



The power-law slope becomes generally steeper with time

III. Total Density Slopes with Redshift



Strong Lensing Observations:

- SL2S:
 - Ruff et al. 2011
 - Sonnenfeld et al. 2013
- SLACS: Auger et al. 2010
- LSD: Treu & Koopmans 2004



Strong Lensing indicates an opposite trend to simulations

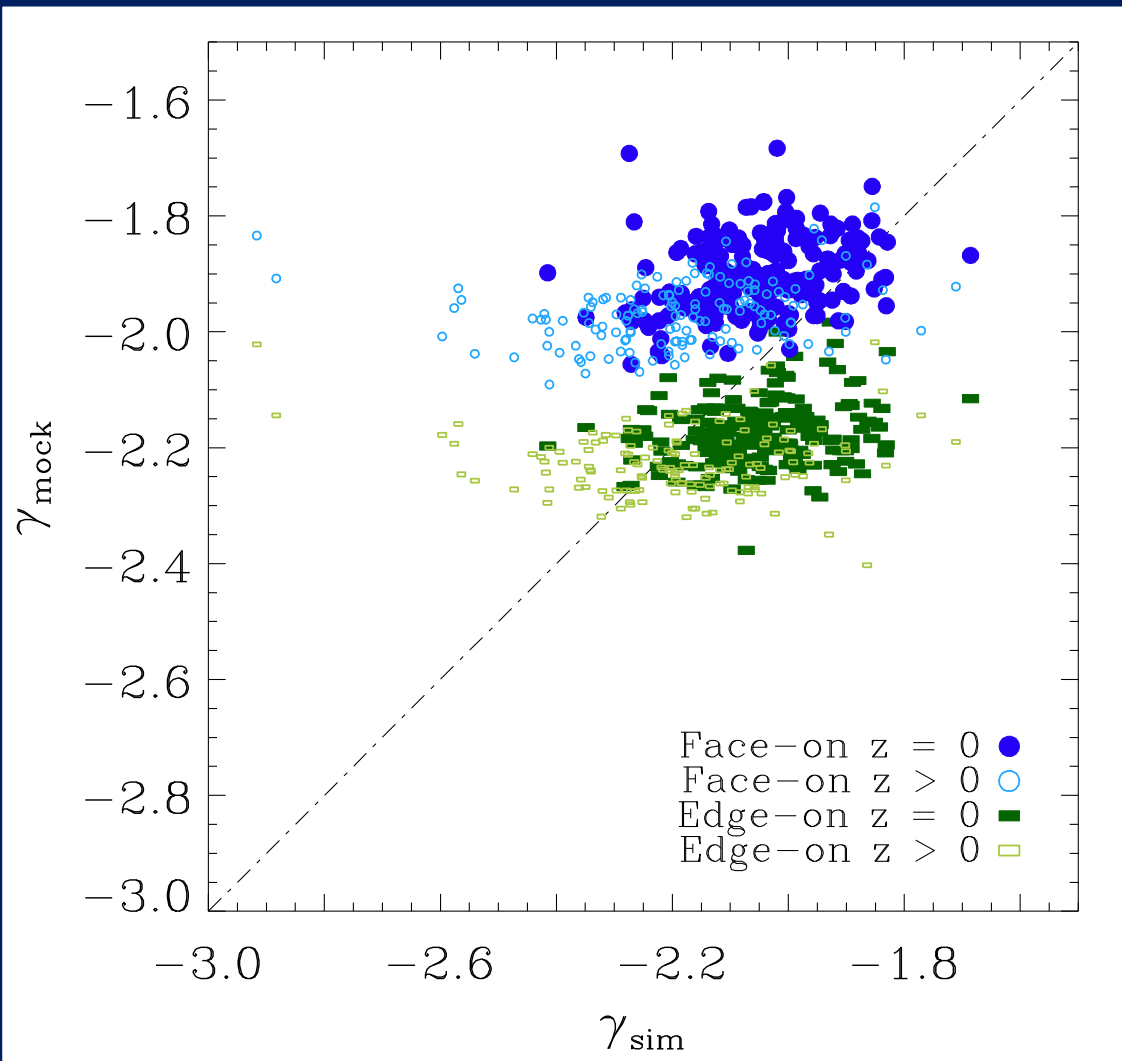
Dynamical Modelling:

- Coma Cluster: Thomas et al. 2007
- SLUGGS: Cappellari et al. 2015



Dynamical Modelling is in good agreement, but only available at low redshift

III. Total Density Slopes with Redshift

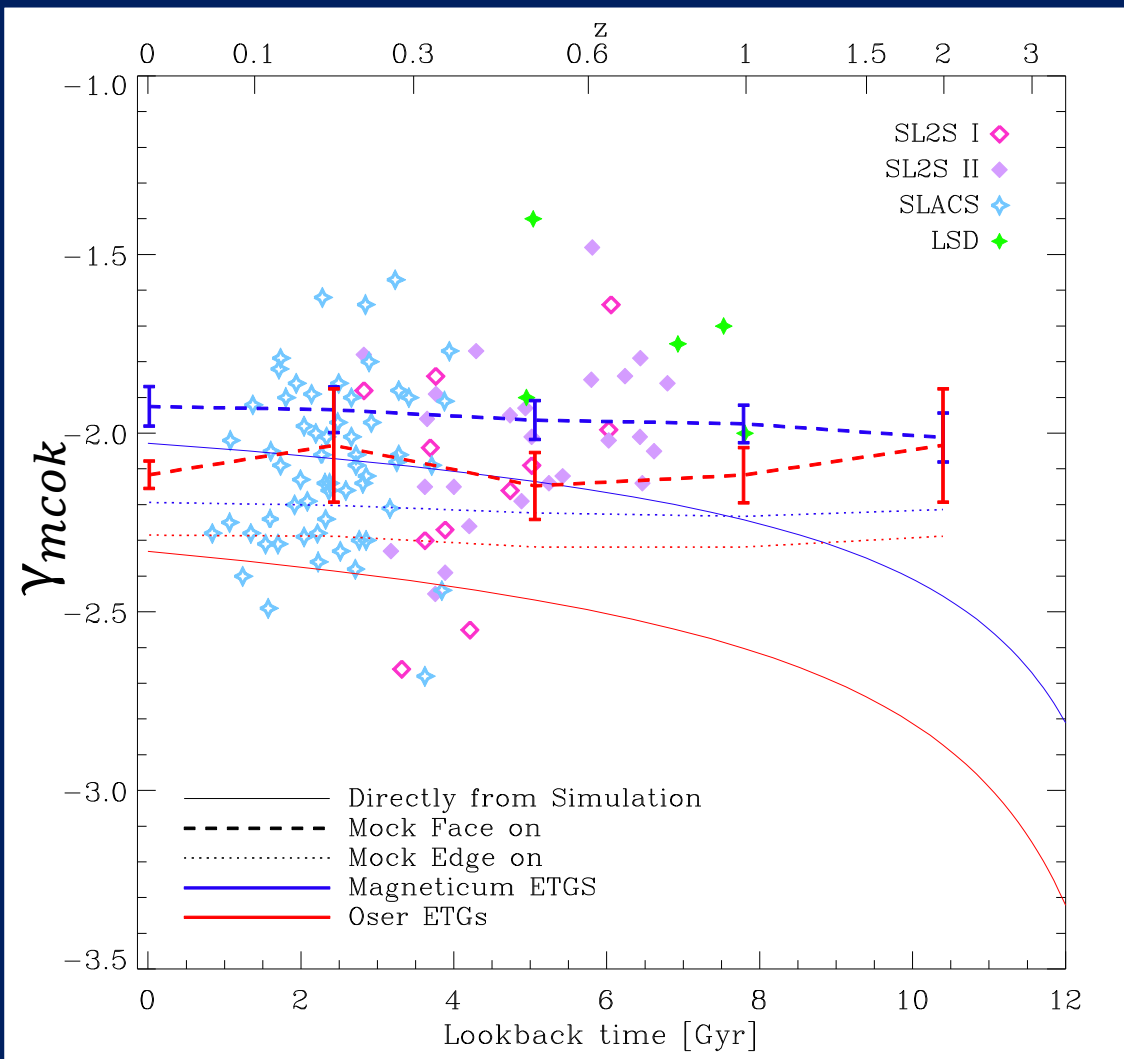


“Mocking” the simulated ETGs:

- $M_{\text{tot}}(R_{\text{Ein}})$
- $R_{\text{eff}} \rightarrow R_{1/2}$
- $R_{\text{Ein}} = 1.5 R_{1/2}$ following Sonnenfeld et al., 2013
- $\sigma_{\text{LOS}}(R_{1/2}/2)$

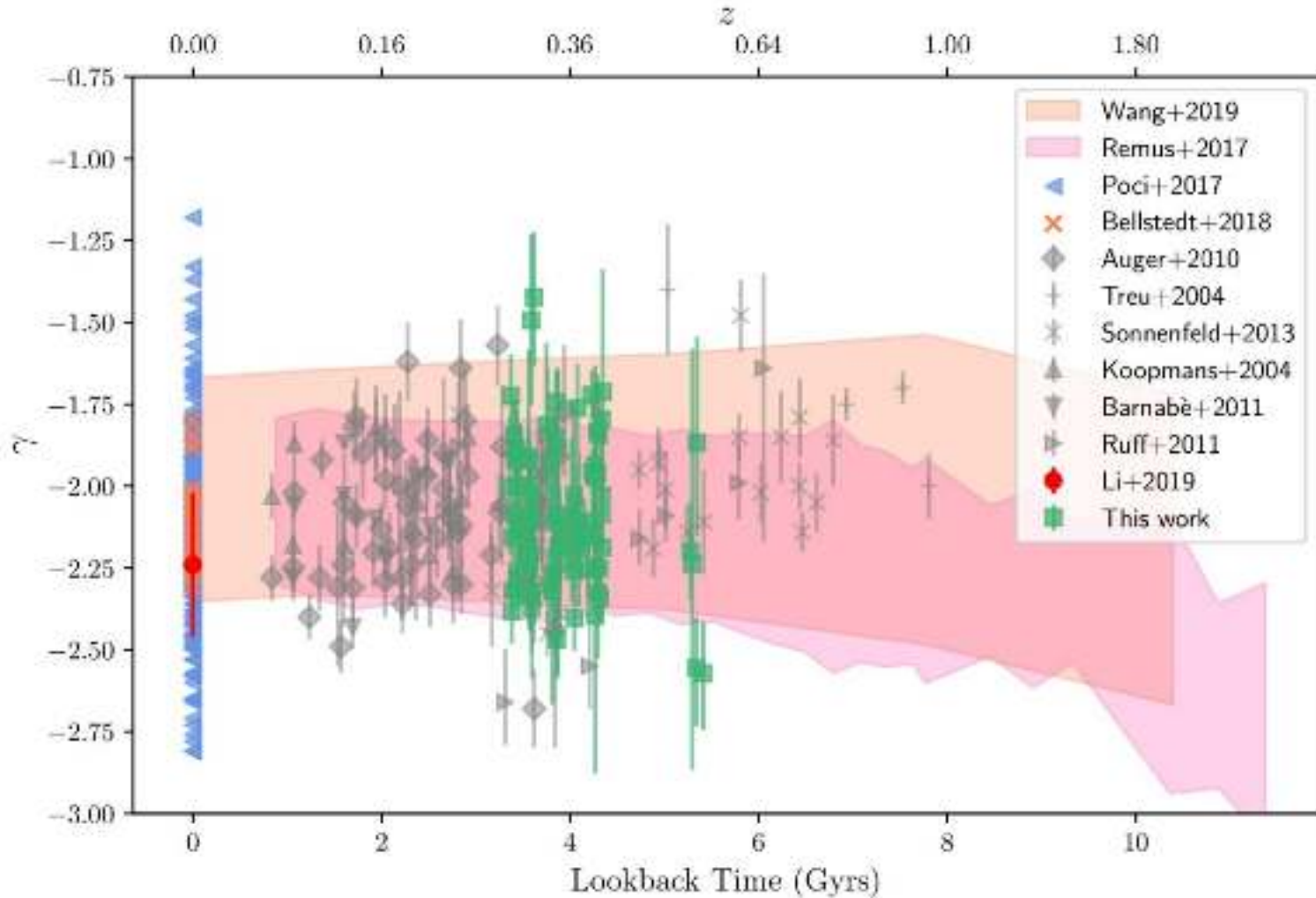
Program kindly provided by A. Sonnenfeld, see
Sonnenfeld et al., 2013, 2015a,b

III. Total Density Slopes with Redshift



We find the same redshift evolution trend as the strong lensing observations if we use the observers' program to mock our simulated ETGs

III. Total Density Slopes with Redshift

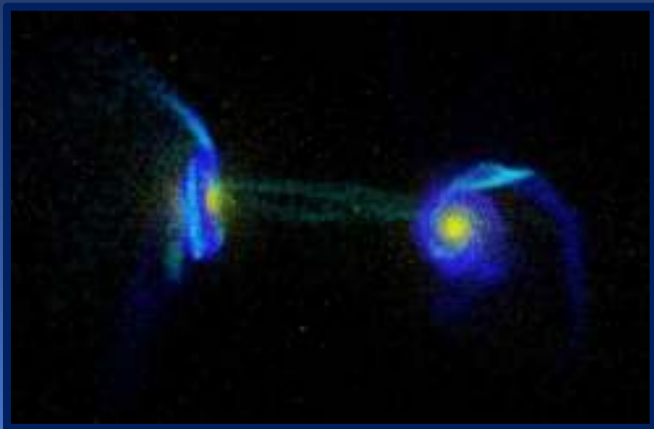


Kinematic measurements
now confirm simulation
trends



Overview: Galaxy Formation Simulations

Isolated Merger Simulations



There are a lot of different idealized merger simulations, but most of them do not have a fancy name

Cosmological Zoom-Simulations



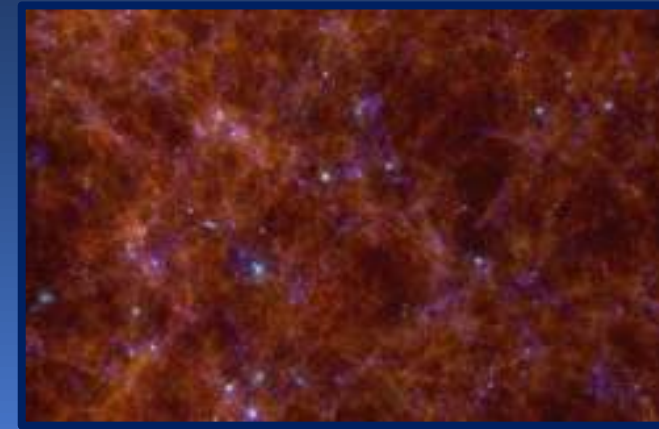
Galaxies (Dwarf to Groups)

- Oser/Naab
- NiHAO
- Fire
- Aquarius
- E-Mosaics

Clusters:

- The 300 Project
- CEagle/Hydrangea
- Romulus-C

Cosmological Box Simulations



Fully Hydrodynamical

- Eagle
- Magneticum
- HorizonAGN
- Illustris/IllustrisTNG
- MassiveBlack

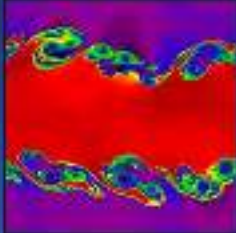
Summary

Smooth Particle Hydrodyn.



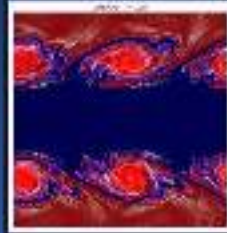
- ✓ Very good conservation properties (mass, momentum, total energy, angular momentum, entropy)
- ✓ shape invariant
- Instabilities do not grow sufficiently
- Mixing behind shocks not sufficient
- Shocks captured by artificial viscosity

Adaptive Mesh Refinement



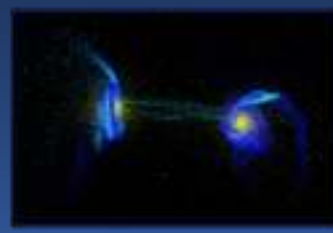
- ✓ Instabilities nicely grow
- ✓ Mixing between phases works well
- Energy conservation issues (especially for fast moving elements)
- Flow over cell boundaries becomes an issue for adaptive meshes
- Not shape invariant

Moving Mesh



- ✓ All good things from the other two
- Flow over cell boundaries (only pseudo-Lagrangian)

Isolated Merger Simulations



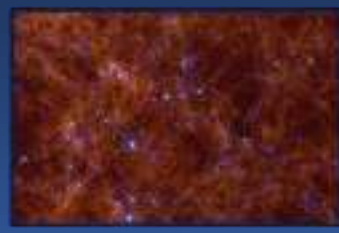
- + high resolution (3000x400x400)
- + controlled conditions
- + parameter study (size, mass, ...)
- no cosmological formation
- not statistical representative
- no redshift evolution

Cosmological Zoom-Simulations



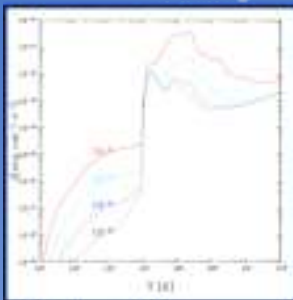
- + high resolution (3000x400x400)
- + cosmological formation
- + target objects (galaxies, clusters, ...)
- biased selection of targets
- not statistical representative
- no parameter control

Cosmological Box Simulations



- + cosmological formation
- + statistical representative
- + no selection bias
- low resolution
- no parameter control
- computationally expensive

Cooling



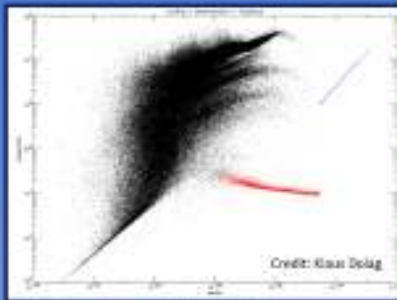
Melo et al., 2007

- Basic Assumptions:
- Optically thin
- Ionization equilibrium (H , H^+ , He , He^+ , He^{++} , e^-)
- 2-body processes ($\propto n^2$)

$$\Lambda(T) n^2$$

BUT: Cooling Catastrophe

Star Formation

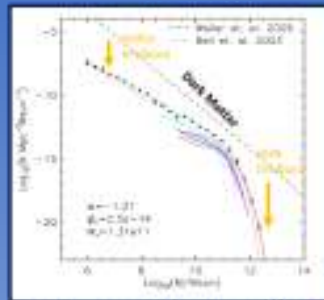


Credit: Klaus Dolag

- Star formation subgrid model:
- Self-regulated star formation
- Set of differential equations needs to be solved.
- Produces reasonable galaxies at low z

BUT: star formation rates at high z not captured

Feedback



Read & Tremblay 2005

- Feedback comes from two different sources:
- Massive Stars and Supernovae
- Supermassive Black Holes (AGN)

Stops the Overcooling Catastrophe

BUT: Burns holes into disks

